claude-windows / c7679509-ee2b-4443-b240-0b3b1b1a7291

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c7679509-ee2b-4443-b240-0b3b1b1a7291\subagents\workflows\wf_a2b97cff-dcd\agent-a7c1983f34e6b9e23.jsonl`
- SHA-256: `eaf0c43c07ca71321de3f0ed9dc71fd97bcbda2f83f972bce4736feb8d472bdb`
- Source modified: `2026-06-21T13:37:25+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_a2b97cff-dcd`
- session_id: `c7679509-ee2b-4443-b240-0b3b1b1a7291`

Transcript

1. user (2026-06-21T13:34:48.460Z)

```
PROJECT: a local, AI-powered badminton/racket-sports HIGHLIGHT INDEXER + rally detector.
Engine repo (PRIMARY): C:/Users/avidu/Projects/badminton-highlight-indexer  (Python, FastAPI +
SQLite + ffmpeg + GPU CV/AI; paid Gemini = cloud AI).
Sibling repos: C:/Users/avidu/Projects/khelsutra (the web/ React SPA + site/ marketing),
C:/Users/avidu/Projects/rally-annotator (VLC plugin, MIT), C:/Users/avidu/Projects/sports-data-
collector, C:/Users/avidu/Projects/sports-obsreport.

GOAL of the plan we are building: bake the engine into a DOWNLOADABLE, "batteries-included"
Python APP that a user can download and run WITHOUT sourcing the repo (no git clone / build-
from-source). It must spin up a local WEB SERVICE that runs (a) INFERENCE (video -> rallies ->
highlight reel), (b) TRAINING (BYO model on their own labelled data), and (c) RALLY ANNOTATION
if needed -- runnable BOTH locally and against CLOUD services. Preserve every already-completed
end-to-end CUJ.

KEY FACTS the orchestrator already verified:
- pyproject.toml: hatchling build, namespace package (no backend/__init__.py),
packages=["backend","training"], console script: indexer = backend.main:main. requires-python
>=3.10.
- torch/torchvision are DELIBERATELY NOT in dependencies -- their CUDA/CPU wheels need --index-
url (PEP621 cannot express). Optional extras: gpu(empty, docs-only), auth(PyJWT),
reporting(git+ssh obsreport), storage-s3(boto3), storage-gcs(google-cloud-storage). google-genai
is a hard dep.
- Server today: python -m backend.main --server --port 8000 . Worker: python -m backend.worker
{infer|train} . CLI single-shot: python -m backend.main --video-path ... --segmenter
{gemini|yolo_hybrid|wasb_hybrid|motion}.
- THE HARD PACKAGING PROBLEM: native WASB shuttle model runs in a SEPARATE Python env (conda,
torch 1.11+cu113) while the main engine uses torch 2.6+cu124. ffmpeg/ffprobe are system deps.
Model weights (wasb_badminton_best.pth.tar) live in a GCS bucket.
- Decoupled-compute SEAM already built (M1-M8 merged, default-OFF): deployment.profile +
compute_target on AppConfig; backend/config/presets.py (PROFILE_PRESETS: truly-local / cloud-
serving / cpu-mock); backend/storage/* (local/gcs/s3/gdrive StorageBackend registry);
backend/compute/* (ComputeBackend registry, CloudRunCompute dispatches a Cloud Run Job);
backend/config/startup.py per-profile checks; backend/billing.py cost ledger.
- The web SPA lives in a DIFFERENT repo (khelsutra/web, Vite/React/TS/Tailwind, i18n en+kn+hi).
The engine has a likely-DEAD backend/api/static. Bundling the UI is a cross-repo concern.
- Licensing guardrail (NON-NEGOTIABLE): every dependency -- code, data, AND weights -- must be
MIT/Apache-2.0/BSD only. Paid LLMs = inference/serving only, never training data. Any NEW
packaging dependency must also be license-clean.
- A prior docs/DESKTOP_APP_PLAN.md exists but was SUPERSEDED by owner decision D10: deliver a
"self-hosted webserver + UI the user spins up" (NOT a desktop app).
khelsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md is the live appliance design.
Always ground claims in file:line or a doc path. Read the actual files; do not assume.
```

AREA: enumerate EVERY completed end-to-end CUJ that the packaged app MUST preserve. READ: the engine memory current-state (C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer/memory/current-state.md) AND khelsutra/docs/DRY_RUN_CUJ.md, FRIENDLY_AND_MULTILINGUAL.md, PER_RALLY_SELECTION.md, and khelsutra memory if reachable. Produce a numbered list of DONE CUJs (e.g. local Gemini run video->12 rallies->reel downloaded en+hi; in-UI gs:// bucket ingest; per-rally selection+query bar; build->compile->download; i18n en/kn/hi auto-negotiation; cold-start GPU native-WASB run; worker infer GCS in/out). For each: a one-line description, its current status (DONE / partial), and the engine+UI surfaces it touches. This list is a hard requirement of the deliverable.

2. assistant (2026-06-21T13:34:54.667Z)

I'll investigate the completed end-to-end CUJs that the packaged app must preserve. Let me start by reading the key documents.

3. user (2026-06-21T13:34:56.092Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-34 of 111 total (47939 tokens, cap 25000). Call Read with offset=35 limit=34 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1       ---
2       name: current-state
3       description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
4       metadata:
5         node_type: memory
6         type: project
7         originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8       ---
9
10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there, don't let them pile up here.
13
14      **?? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session):** SPA is fully **en+kn+hi (DONE)**; a real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical engine bug fixed** (#281). **? NEXT BUILD (owner-picked): (1) Path B ? point at a bucket video IN THE UI.** ? **B-P1 VERIFIED ALREADY DONE ? NO ENGINE CODE (do NOT re-scope it as a feature):** `POST /api/process` has accepted `gs://`/`gcs://` URIs since **M2/#280** (submit resolves the URI + skips the local-exists-404 for non-local backends `main.py:283`; the worker fetches to scratch `acquire_local_input main.py:159` ? same hash?validate?segment?DB pipeline ? queryable API-DB `video_id`); **TESTED** (`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` + hosted-mode-reject / unknown-scheme-400 / CLI variants). Confirmed by a **3-skeptic verification workflow (0 refuted, gaps:[])**. **Scope pin:** holds in DEFAULT single-user mode (hosted-mode `allowed_video_root` correctly **400s** a bucket URI ? by design). Stale "local-path-only / net-new" docs **CORRECTED ? khelsutra [#55]** (DRY_RUN_CUJ §1/2/4/6/8 + LOCAL_APPLIANCE + README). **? So Path B = ONLY B-P2 = the SPA ingest/progress screen** (enter a `gs://` URI ? `POST /api/process` ? poll `/api/jobs` ? load `result.video_id`; back-half already works) ? **100% khelsutra `web/` SPA + i18n (en/kn/hi), ZERO engine code.** Ready B-P2 sketch: `DRY_RUN_CUJ.md` §2 (new `web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` + `App.tsx` videoId?state handoff + `ingest.*` keys). See [[lesson-verify-engine-surface-before-scoping]]. **(2) Droplet deploy (opt 3):** deploy the FIXED engine to `168.144.126.176` (un-provisioned, 2-core/3.8GB) ? re-run CUJ on Linux. **STILL OPEN (localization DoD):** B3 (`site/` marketing still English-only) + native-speaker review of the AI-drafted kn/hi. **Ramp-up:** `khelsutra/docs/DRY_RUN_CUJ.md` + `FRIENDLY_AND_MULTILINGUAL.md` §7 + the dry-run checkpoint below. **Clean slate (at session start):** both repos on master, 0 open PRs, no worktrees/VMs. khelsutra master `2421557` · engine master `2afd54f` (? origin/master `cee4baa`, #283 M4 squashed).
15
16      **Checkpoint:** 2026-06-21 (ENGINE SERVING M5+ ? OWNER BATCH-AUTHORIZED the independent
```

path). Owner redirected: "first surface the owner-gated decisions so I can take them + unblock
max independent PRs." Ran a 6-agent grounded gate-map **workflow** (`wf_4849400f`) over README
§2/§7/§8 + code ? a decision menu splitting every M5?M12 gate into **(a) default-OFF code I can
merge now** vs **(b) thin owner residual**. **Owner answers (AskUserQuestion):** **(1) BLANKET
YES** ? write+merge ALL default-OFF serving code (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each; **+ SPEND OK up to $100/?10k** for due-diligence +
production-grade testing/verification (so the real M6/M7 cloud runs are ALSO unblocked
within the cap ? must respect it, DELETE resources after, confirm cmd+cost before each spend per
[[gcp-vm-always-delete]]). **(2) MERGE each on green CI** (OAuth real-testing deferred to a
later rigorous pass). **(3) ALL recommended design picks** "with generous budget in mind": **co-
located stitch** (D4) · **libx264 ENCODE + NVDEC DECODE-only** (protect parity) · **hard-reject
`<2GB` v1** (config knob) · confirm `COMPUTE_DECOUPLED_SERVING/` name · **M6 = Cloud Run JOBS +
bucket-poll completion**. **? PLANNED PR ORDER:** A(doc: §8 resolutions + P3 reconcile + tracker
housekeeping) ? B(M5 preset+startup-checks+L1?L4 parity test) ? C(M6 `backend/compute/` registry
+ Dockerfile + mocked-dispatch) ? D(M8 v6 migration + pricebook[$0 placeholder] + USD/INR
aggregation + /cost API) ? E(M9-auth Cf-Access JWT verify + tenant scoping; M9-consent inert
NULL cols EXTEND D's v6 migration) ? F(M11 capture?store?shadow-eval?goldenize plumbing,
`flywheel.*`=OFF) ? G(M12 `GDriveApiStorage`+`gdrive-api://`, fake-Drive tests) + Ops-Agent
codify into `deploy/gcp/provision.sh`. Coordinate: D/E/M9-consent share ONE v6 migration (author
in D, extend in E). **Genuinely owner-only residual:** real GPU run (M5 runlog) [or I run on
cloud GPU within cap], counsel ToS/DPA (M9-consent text), Google OAuth app (M12 real), real GCP
prices+FX (M8 calibrate), CF team-domain/AUD+ingress-lock, repo rename, DO droplet/token retire.
? shared checkout ? sibling merged **#284** (`06ff4b7`, bucket-URI regression guard) mid/just-
before this session; RE-FETCH origin/master before each branch. **? PROGRESS (this session):** ?
PR-A [#285] MERGED (`fefa412`, doc: §8 D16/D17 resolutions + P3 reconcile + tracker
housekeeping). ? **M5 [#286] MERGED** (`b5af66e`, all 7 CI green) ? `backend/config/startup.py`
per-profile fail/warn-fast checks (cloud-serving+non-cloud-storage = the one fatal coherence
error; GPU/creds = warnings) wired into server (`_enforce_startup_or_exit`?exit1) + worker
(`cmd_infer`/`cmd_train`?rc2), **default-OFF + probe gated behind active profile (worker stays
torch-free at profile='')**; **L4 profile-parity test** (`tests/test_profile_parity.py`: same
stub trajectory?byte-identical windows+ranges across truly-local/cpu-mock/no-profile + worker-
vs-in-process); 4-lens adversarial review folded (`wf_34671ac1`: MAJOR torch-free fix + minors).
Runlog template pre-staged; the real-GPU truly-local **runlog = owner L5**. M5 ? in §7 tracker.
? **M6 [#287] MERGED** (`8cdf65c`, all 7 CI green) ? `backend/compute/` ComputeBackend registry:
`get_compute_backend(config)` returns a backend ONLY for `compute_target=cloud_run` (else `None`
? worker's in-process path, **default-OFF**); `CloudRunCompute` dispatches a Cloud Run **Job**
(injectable `CloudRunJobClient`; real `GoogleCloudRunJobClient` lazy google-cloud-run) ?
bucket-polls a done-marker via `execution_policy.poll_until`+storage (D16); container forced
inline (`compute_target=local_gpu`+native, never re-dispatch); worker `--done-uri` writes a
success marker. `deploy/cloudrun/{Dockerfile,README}` = L4 image (CUDA+ffmpeg+DejaVu
fonts+`wasb` conda torch-1.11 env mirroring gpu_setup.sh; weights staged per WASB-C3) + M7
runbook. 4-lens review (`wf_87b22fa4`) folded a **BLOCKER** (cloud `fetch` needs a dst ? reads
silently returned `{}` on gcs/s3; tests passed only b/c local) + poll-wait test gap +
`PolicyTimeout`?clean rc 4. Suite 1167 green. ? **M8 [#288] MERGED** (`43eccc1`, 7 CI green) ?
`backend/billing.py` (pure `usage×pricebook`, USD source-of-truth / INR display) + **v6
migration** (NULLABLE cost cols on `jobs` + versioned `pricebook` seeded `placeholder-v0` at $0;
additive-NULL, idempotent) + `record_job_cost`/`get_job_cost`/`get_video_cost` + `GET
/api/jobs|videos/{id}/cost` + `BillingConfig` (**default-OFF**) + gated
`_maybe_record_job_cost`. 3-lens review (`wf_498b24af`) folded a **MAJOR silent-GPU-under-bill**
(an unpriced `gpu_type` is now SURFACED ? WARNING + `unpriced_gpu_seconds` breakdown line, not
invisible $0) + the **honest no-op** caveat (pipeline doesn't emit `result['usage']` yet ? cost
0 until a follow-up wires the metering; documented + tested) + margin/rounding reconciliation.
Suite 1186 green. **?? 4 SERVING PRs THIS SESSION, ALL MERGED: #285(docs §8 D16/D17 + P3
reconcile) + #286(M5 startup-checks+L4 parity) + #287(M6 Cloud Run dispatch shim+image) +
#288(M8 cost ledger).** Each default-OFF, isolated, adversarial-reviewed, full-suite+CI green,
tracker+changelog in-PR. **? v6 is now SHIPPED (M8 took it) ? the NEXT migration is v7**
(M9-consent cols + per-video P1 `workspace_uri` ? the earlier "extend v6" plan shifts to a fresh
v7 `_migrate_to_v7`). **? REMAINING authorized batch (ready to build next, blueprints loaded):**
M9-auth = add `PyJWT[crypto]`; a FastAPI dep extracts `Cf-Access-Jwt-Assertion` ?
fetch+cache the team JWKS ? verify sig/exp/iss/aud ? `user_id`; per-tenant scoping on the v4
`user_id` cols; tighten the placeholder CORS (`main.py:40`); all behind
`security.edge_auth_enabled=False`; offline RSA/fake-JWKS tests. **M9-consent** = inert NULL
`consent_status/is_adult_attested/retention_class` in a **v7** migration (safe no-train default)
? the ToS/DPA TEXT is owner/counsel. **M11** = `flywheel.capture/shadow_eval/promote_stub`
reusing `workspace.py::for_video` + `backend/eval/experiment.compare`(:485) +

`promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked `consent_attested`.
M12 = `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme against an INJECTED fake
Drive (mirror `gcs.py` DI), keep the mount-only `gdrive://` (truly-local) untouched. **Ops-Agent
codify** into `deploy/gcp/provision.sh` (logging+monitoring APIs + SA metric/log-writer roles +
startup-script ? [[gcp-ops-agent-enable]]). **? owner-only residual:** real GPU run ? M5 truly-
local runlog (template pre-staged in `runlogs/`); GCP spend for the real M6/M7 run within the
$100/?10k cap (build the image per `deploy/cloudrun/README`, run on L4, capture
DEPLOY_REPORT, **DELETE after**); counsel ToS/DPA (M9-consent); real GCP prices+FX + the M8
usage-emission follow-up; Google OAuth app (M12 real); CF team-domain/AUD + ingress-lock; repo
rename; DO droplet/token retire. **?? REAL M7 CLOUD GPU RUN ? ? DONE + VERIFIED (owner:
"complete any and every cloud run with GPUs").** Provisioned an **L4** `rally-m7-gpu`
(`g2-standard-4`, **asia-southeast1-a** Singapore ? both Mumbai zones STOCKOUT for L4; reads the
Mumbai bucket cross-region) ? `bootstrap --gpu` + `gpu_setup.sh` (wasb conda env, **torch
1.11+cu113 cuda True on L4** via PTX-JIT) + staged the 5.82 MiB weights ? `worker -v infer
--video-uri gcs://?/mahadevpura_smoke_180s.mp4 --out-uri gcs://? --segmenter wasb_hybrid` ? **22
rallies** (5395 trajectory points, 2930 visible; Phase 2 WASB 363.7s) ? pushed `gs://khelsutra-
rally-corpus/outputs/05e0e577?/{intervals.csv,report.json}` (segment_count 22, status success).
VM DELETED, audit 0 instances/0 disks ? $0 (~18 min ? **$0.21**, well under the cap). **?
SOLVED the long-flagged "cold-start gives 0 rallies" mystery ? THE MISSING PIECE = `--set
indexing.skip_ai_handoff=true`:** `wasb_hybrid` does a Phase-3 **Gemini handoff** by default;
with no `GEMINI_API_KEY` on the box it errors ? 0 segments. `skip_ai_handoff=true` = the
fully-local path ? emits the 22 WASB candidate windows directly as rallies (no Gemini). So a
no-Gemini cloud run MUST set `skip_ai_handoff=true` (this is exactly M5's truly-local preset,
which sets it). ? minor: `-v` is a TOP-LEVEL flag ? `worker -v infer ?` (not `infer -v`).
DEPLOY_REPORT runlog = TODO (after M9-auth). **? NOW CONTINUING (owner: "keep going with auth
and other work after that"):** building **M9-auth** (branch `feat/serving-m9-auth` off
`43eccc1`) ? `backend/api/auth.py` Cf-Access JWT verify (lazy `PyJWT[crypto]`, injectable JWKS
for offline tests) +
`SecurityConfig.{edge_auth_enabled=False,cf_team_domain,cf_access_aud,cors_allow_origins}` +
configurable CORS + owner_id tenant tag; then M11/M12/Ops-Agent. [[gcp-vm-always-delete]]
[[working-agreement-merging]] [[feedback-launch-quality-bar]] [[compute-stack-gcp-only]]
17
18 **Checkpoint:** 2026-06-21 (STALENESS-FIX / Path-B verify) ? owner: pick up Path B +
"fix all stale docs and stop next sessions spiralling into the staleness." **? Verified-FIRST
finding: B-P1 was ALREADY DONE** ? `POST /api/process` accepts `gs://`/`gcs://` since
M2/#280 (resolve?skip-exists-404?worker fetch?same pipeline?queryable API-DB `video_id`) +
TESTED; confirmed by inline file:line trace AND a **3-skeptic verification workflow (0/3
refuted, gaps:[])**. So the doc-scoped "build the gcs ingest endpoint" was a phantom; Path B =
SPA-only (B-P2). **DONE this session:** (1) **khelsutra [#55] MERGED** (master `d0461fd`) ?
corrected `DRY_RUN_CUJ` §1/2/4/6/§8 + `LOCAL_APPLIANCE` (`/api/health` EXISTS now M1/#268;
bucket-URI ingest NOT a net-new endpoint; T12 engine-half ? done) + `README`; added the anti-
spiral convention *"re-verify engine-surface `[gap]`/`net-new` claims at a cited file:line ? the
engine ships PRs between sessions."* (2) **Memory steering corrected** (this NEXT-SESSION-FOCUS
block + the dry-run checkpoint were stale "local-path-only") + new lesson [[lesson-verify-
engine-surface-before-scoping]] + MEMORY.md index. (3) **engine [#284] MERGED** (master
`06ff4b7`, all CI green incl. full suite 2m32s) ? regression GUARD
`test_process_bucket_uri_yields_queryable_video_id` (parametrized gs://+gcs://) pinning
`video_id==MD5(fetched bytes)` (not the URI) + `POST /api/query` queryability (the SPA contract;
previously only LOCAL had it); built+removed in an isolated worktree off `origin/master` (engine
checkout left on its concurrent-session branch). **? NEXT: B-P2** = the SPA ingest/progress
screen (enter gs:// URI ? /api/process ? poll /api/jobs ? load video_id; back-half already
works) ? **100% `web/` + i18n (en/kn/hi), ZERO engine code**; ready sketch in `DRY_RUN_CUJ.md`
§2 + the workflow synthesis
(`web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` + `App.tsx`
videoId?state + `ingest.*` keys). Then opt-3 droplet (deploy fixed engine + GCS creds + #281 fix
? re-run CUJ on Linux). **? Repos SYNCED TO HEAD at session end:** khelsutra master `d0461fd`;
engine working tree `06ff4b7` ? NOTE the engine main checkout couldn't switch to `master`
(sibling worktree `tier1-windowing` holds it), so it's the merged-but-remote-gone
`feat/serving-m4-device-context` branch **reset --hard to origin/master + upstream repointed to
origin/master** (safe: tracked tree clean + the 2 divergent commits were the M4 #283 squash,
byte-identical to master; old tip tagged `_pre_sync_m4_44c9f01` + reflog). So `git pull --ff-
only` works there now; the branch NAME is cosmetic (points at master head).
19 - **? B-P2 BUILT (in-UI bucket ingest) ? khelsutra
[#56](https://github.com/avidullu/khelsutra/pull/56) OPEN, CI GREEN (web+site+CF build),
awaiting owner merge** (feature PR ? left for owner; precedent exists for me merging SPA PRs on

owner's "merge as you go"). 100% `web/` + i18n, ZERO engine code (rides B-P1). New `web/src/{api
/types,api/client(processVideo/getJob),lib/errors(parseDetail+classifyError),hooks/useIngestJob(
idle?submitting?polling?done|error; recursive-setTimeout poll backoff+10min-ceiling+1 cid;
success-shaped-failure?emptyResult),components/IngestPanel}` + `App.tsx` videoId
useMemo?useState + IngestPanel-when-!videoId + history.replaceState; `ingest.*` en/kn/hi (AI-
draft, native-review-pending). **123 web tests (+19), tsc/build green, browser-verified** (panel
renders en+hi; live submit?friendly localized "is the engine running?"; cid logs). Live-fixed
UX: bare 5xx/proxy-down ? friendly `networkError`, not "HTTP 500". `DRY_RUN_CUJ §6` box3 = ?
(flips on a live **e2e-from-UI** vs a real engine). **? NEXT:** merge #56 ? live e2e-from-UI
(local or droplet opt-3) ? flip box3; then localization DoD tail (B3 site `t()` shim; native
kn/hi review).
20
21 **Checkpoint:** 2026-06-21 (NEW SESSION ? UX+l10n focus STARTED) ? ramped up via a
research+verification **workflow** (web/+site/ string maps, Indic-l10n prior art, **5
adversarial verdicts:** both surfaces **ZERO i18n today**; recording-edu **already shipped** on
the homepage gist + `/capture`; `I18N_PLAN` targets the **dead** `backend/api/static` so its
P1/P3 wiring retargets to site/+web/). **? Isolated the work into ONE umbrella tracked project ?
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`** (A: non-tech UX + B: kn/hi l10n; §7 P-tracker =
source of truth; **archive when en+kn+hi render on BOTH surfaces**). Pointers wired: khelsutra
`docs/README`+`NEXT_STEPS`; engine `I18N_PLAN`+`UI_PROPOSAL` banners + `DOC_STATUS` §3a/§3b rows
(engine edits in an **isolated worktree off origin/master** ? engine checkout is a concurrent
session's branch; their findings-#1/#2 PR **#274 merged mid-session** ? master `03fa951`).
**Owner (AskUserQuestion) locked D6 (en+kn?hi) · D7 (OFL Noto self-host) · D8 (SPA-first) · D9
(catalogs in khelsutra); only D-BACKEND-DEPTH deferred. 3 PRs OPEN:** khelsutra docs
[#41](https://github.com/avidullu/khelsutra/pull/41); engine pointers
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275); **B1 SPA i18n
scaffold + language switcher (en-only) ? [#42](https://github.com/avidullu/khelsutra/pull/42) ?
CI GREEN (web+site pass), verified LIVE in-browser** (switcher EN|????? persists; content falls
back to en with no kn catalog; active-highlight follows the *chosen* locale ? a bug caught by
browser verification + fixed). B1 = react-i18next + ICU + detector + `en` catalog + the SPA's
FIRST jsdom render tests (79 tests green); deps MIT/BSD-clean. Engine pointer PR built in an
isolated worktree (now removed); engine checkout kept flipping branches (concurrent session
active) ? never disturbed. **? NEXT slices (tracker §7): B2** (full ~50-string extraction + key-
parity/no-bare-strings CI) ? **B3** (site `t()` shim) ? **B4/B5** (Noto fonts + kn catalog via
Gemini-draft?review). **Also (owner ask, 2026-06-21): the en/kn/hi academy flyers already
existed** (a sibling merged PR **#40** ? master `37053b9`; flyer.html / flyer-kn.html / flyer-
hi.html served at /flyer{,-kn,-hi}) but were UNLINKED ? **linked all three from the homepage
footer + added VM-host clean routes ? [#43](https://github.com/avidullu/khelsutra/pull/43)
(MERGED ? master `f121c7f`).** Then **owner asked for MORE flyer placement ?
[#44](https://github.com/avidullu/khelsutra/pull/44): "Flyer" in the desktop top-nav + the
3-language links on own-model's foot** (44 site tests green; verified live). (Flyers use Google-
Fonts Noto ? separate from the app's D7 self-host; not refactored.) **Owner #41 comment ? added
a rigorous §6.5 "automatic language-negotiation" testing strategy** (matrix N1?N9: a kn/hi
BROWSER auto-gets the respective version; DoD gate) **+ concrete browser-language detection
tests in B1 [#42]** (N1/N3?N7; 84 web tests green). #41/#42/#275/#44 all merged. **?? THEN
owner: "take it to completion until you can run a screencast with a kannada-default browser ?
right page renders ? switch hi ? en; merge PRs as you go." ? ? DONE for the `web/` SPA.** Merged
in sequence (I merged each on green CI, owner-authorized): **B2 [#45]** (full ~50-string
extraction + key-parity/no-bare-strings CI) · **B4 [#46]** (self-host OFL Noto
Kannada/Devanagari, @fontsource-variable) · **B5 [#47]** (Kannada catalog) · **B6 [#48]** (Hindi
catalog ? the extensibility proof: just a catalog + supported-list entry). **SPA now fully
en+kn+hi; 99 web tests green** (incl. N1/N2 browser-detection, B5/B6 render, key-parity
kn==hi==en, `<html lang>` sync). **? SCREENCAST PRODUCED** (Playwright `locale: kn-IN` ? Kannada
auto-renders ? switch hi ? en; `C:\Users\avidu\Projects\khelsutra-
screencast\khelsutra-i18n-screencast.mp4` (9s) + per-locale frames). Tracker milestone ? [#49].
? **kn/hi are AI-DRAFTED (me) ? native-speaker review pending** before wide public launch. **?
REMAINING for full §9 DoD (NOT archive-ready):** **B3** (the `site/` marketing surface ?
homepage/capture still English-only ? via the shared `t()` shim) · **A3** recording-education
kn/hi · the **non-tech UX A-items** (A1/A2/A4/A6...) · native review. Playwright installed in
isolated `C:\Users\avidu\Projects\khelsutra-screencast\` (kept out of web/ deps). **THEN owner
polish batch (2026-06-21) ? all merged:** (1) **OFFLINE/self-contained verified + guarded ?
[#50]:** the SPA is air-gap-renderable (Noto = local woff2 via @fontsource, `@font-face`
url?local `/assets`, no Google-Fonts/CDN, `web/index.html` no external link); CI guard added so
it can't regress. *(The marketing `site/` still uses Google-Fonts ? hosted/online surface, not
the offline tool; self-host = optional follow-up.)* (2) **SCREENCAST uploaded ? `gs://khelsutra-

rally-corpus/promo/khelsutra-i18n-screencast.mp4`** (+ local). (3) **Localized SAMPLE-QUERY PROMPT (en/kn/hi) ? [#51]:** `query.hint` = instructive "type `over 10s` for rallies >10s ? or tap a sample" + `query.try` labels pills "Sample queries"; verified live kn. (4) **§6.5 testing rule:** i18n testing is now a MANDATORY merge-gate for any future query-parsing/semantic-understanding/NL work (parser is English-grammar-only today, A9). **103 web tests green.** NO open PRs. [[next-session-ux-localization]] [[doc-status-register]] [[working-agreement-tracked-project-docs]]

Checkpoint: 2026-06-21 (DRY-RUN CUJ + a CRITICAL ENGINE FIX) ? owner: "stitch the UI to the tool; dry-run a **GCS-bucket video** via the UI ? **reel downloaded local**; repeat in Hindi. Run locally (opt 1) then droplet (opt 3); report both; identify fixes + merge." Chose **A-now-then-B**, **Gemini**. **Tracked goal ? `khelsutra/docs/DRY_RUN_CUJ.md`** ([#53]/[#54]). **? PATH-A LOCAL RUN DONE (en + hi):** isolated engine off origin/master (fresh DB/output worktree) + `.env` Gemini key ? `gsutil cp gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` ? `POST /api/process {gemini}` ? **12 rallies** (`video_id 05e0e577?`) ? SPA `npm run dev` (Vite proxies /api?:8000) `?video=05e0e577?` ? loaded 12 real rallies ? **Build ? /api/compile ? reel (1080p, 76.5s) DOWNLOADED to `~/Downloads/khelsutra-dryrun/highlights_local_{en,hi}.mp4`** (50.9MB). **Hindi repeat ?** (page/cards/"??? ?????"/"???? ??? ????? ??" all Hindi; reel downloaded). **?? CRITICAL BUG FOUND + FIXED ? engine [#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281) MERGED:** running the documented `python -m backend.main --server` runs main.py as `__main__` (sets `__main__.db_instance`), but `job_worker`/`uploads` read `import backend.main` ? a SEPARATE module whose `db_instance` stays None ? the async-job drain crashes silently on `None.claim_due_job()` ? **every /api/process job hangs in `queued` forever**. Latent: unit tests monkeypatch the executor inline; the UI was verified vs a STUB engine. Fix = dispatch `__main__`?`backend.main`; + `tests/test_server_entrypoint.py` (real subprocess guard, FAILS-without/PASSES-with). Full engine suite green (2m23s). **Unblocked the run via the single-module invocation** (`python -c "...; from backend import main as m; m.main()"`). ? **OPT-3 DROPLET (168.144.126.176) NOT YET RUN:** reachable (Linux, py3.12, ffmpeg, 2-core/3.8GB) but **un-provisioned** (no `~/rally` engine checkout) ? needs a from-scratch engine install OR a proper deploy of the FIXED engine (the fix applies there too). **? NEXT:** opt-3 droplet run (deploy fixed engine ? re-run CUJ); Path B (in-UI bucket ingest). ? **CORRECTION (next-session verify): the "/api/process is local-path-only / net-new gcs ingest" note here was STALE** ? `/api/process` already accepts `gs://` (M2/#280, tested); Path B is **SPA-only (B-P2)**, no engine endpoint. (The "worker writes CSV to the bucket not the API DB" bit is the *separate* `worker infer` CLI, not the API path ? that's what the stale claim conflated.) Docs fixed ? khelsutra [#55]; see the NEXT SESSION FOCUS block above + [[lesson-verify-engine-surface-before-scoping]]. [[monetization-plan]]

? HANDOFF (2026-06-20) ? KHELSUTRA.GURU brand site: shipped, live, portable. Resume the product threads below. Separate track from the indexer rally-detection handoff that follows. Full durable record + ramp-up: [[khelsutra-domain-launch]]. Repo = **`avidullu/khelsutra`** (NOT this indexer repo).

- **? START HERE (2026-06-20, latest ? big session, ALL MERGED, NO open PRs).** Per-rally **MVP/M0 COMPLETE** + a full product/promo wave. **khelsutra master `a7214ad`** · **engine (Khelsutra/badminton-highlight-indexer) master has M1 [#268]**.
  - **Per-rally MVP (khelsutra `web/` SPA):** P3 `parseQuery()` query bar [#28] (length/order/count/shot_count; greys shot-type/player/score "not available yet" ? never faked; **13 adversarial-review findings folded in**) · P4 `SelectionFooter`+honest `min_shot_count` warning [#30] (`web/src/lib/floor.ts` mirrors `main.py:362-383`; no-shot_count exempt; all-below ? Build DISABLED) · P5 real load `?video=<id>` + Build?`compileReel`?`<video>` preview+download [#31]. Full obs (`query.*`/`build.*`/one cid). **Then: lead-with-LENGTH, shot-count flagged EXPERIMENTAL** [#33] (cards show `m:ss long`; amber "experimental" chip; duration-led examples; badge `P5`?`alpha`; parseQuery still accepts shot-count, de-featured). Suite **75 green**.
  - **Engine M1 [#268] (GCP/org repo, built in an ISOLATED git worktree off origin/master so the concurrent session's checkout was untouched):** `GET /api/health` ({status,sport,default_segmenter}) + `POST /api/compile` now returns **`download_url`** (additive; `filepath` unchanged). Full engine suite **1019 passed/7 skipped**, ruff+mypy clean, CI green (coverage floor incl. torch). *(SPA can adopt `download_url`/offline-banner as a follow-up.)*
  - **GCS round-trip (reuse artifacts, CPU/no-GPU):** pulled `mahadevpura_smoke_180s.mp4` + intervals from the bucket ? **stitched a curated top-10-longest highlight** (ffmpeg) ? **put it back** at `gs://khelsutra-rally-corpus/highlights/curated_top10_longest.mp4` (61s).
  - **3 screencasts (Playwright?webm?ffmpeg):** `promo_cuj.mp4` (old UI, real footage) ·

`full_cuj_real.mp4` (NEW duration-first UI + REAL curated reel ? the definitive full-CUJ) ? both in `gs://khelsutra-rally-corpus/promo/` (real footage ? bucket, NOT public repo) · **`site/public/video/tool-cuj.mp4`** (NEW UI, **footage-free** branded-placeholder reel ? safe to self-host). Local copies in `badminton-highlight-indexer/artifacts/promo/` (gitignored).
31   - **Site (excite-to-join) [#34/#35]:** after sign-up the reveal now shows **(1) the tool in action** (self-hosted footage-free `tool-cuj.mp4`) **+ (2) a ~30s reel preview** (`REEL_SECS` 90?30, gated nocookie YouTube ? real faces stay in the unlisted reel, not the repo); `revealAfterSignup`; coherent CTA/teaser copy; `server.js` serves `.mp4` as `video/mp4`; `*.mp4`/`*.gif` marked binary. Site suite **34 green**; **droplet re-synced + parity GREEN** (index+mp4 byte-identical CF?VM:8787).
32   - **? GPU note:** could NOT spin a fresh GPU VM ? `GPUS_ALL_REGIONS=1` slot held by the concurrent session's `rally-gpu` M2 training (did NOT disturb). Validated CUJ against the first-L4-run REAL artifacts instead. A fresh live `/api/process` GPU inference stays **deferred** until the GPU frees (low-risk existing path). **Owner observation ? [[machine-pooling-idea]]** (pool GPUs vs spin/kill per task).
33   - **Owner decision flags (PUBLIC promo):** `full_cuj_real.mp4`/`promo_cuj.mp4` show real court footage (faces) ? for public use beyond the gated site, confirm. The footage-free `tool-cuj.mp4` is safe.
34   - **? COLD-START RUN (2026-06-21, owner-approved; GPU freed ? the other session deleted `rally-gpu`):** provisioned a FRESH **L4** `rally-coldstart` (T4 stocked-out asia-south1-a ? L4 -c), built engine+native-WASB env in **~7 min** (`bootstrap --gpu`+`gpu_setup.sh`+weights; main torch 2.6+cu124 & wasb-env 1.11+cu113 both cuda True), `worker infer` GCS-in?GCS-out, **VM DELETED ($0, ~$0.40 total)**. **? Cold-start MECHANISM fully validated end-to-end.** **? BLOCKER: native WASB returned 0 rallies** on mahadevpura-1 (full-res), GX030094_proxy, AND the known-good **smoke clip (gave 22 in the first L4 run)** ? a **fresh-build CONFIG/wiring issue, NOT the CUJ**: my minimal reconstructed `config.json` (`indexing.detector_impl=native`, `default_segmenter=wasb_hybrid`, no extra fusion/gating tuning) doesn't reproduce the engine session's TUNED `worker infer detector_impl=native` invocation. **? THE missing piece for a repeatable cold-start = the engine session's EXACT native-WASB config/cmd** (the one that gave 22 rallies). Side finding: `mahadevpura-1.MP4` is full 1080p/~25Mbps; the working clips are downscaled `*_proxy` ? feed proxies (or add a downscale step). ? The diagnostic infer **clobbered** `outputs/05e0e577?/{intervals.csv,report.json}` ? **RESTORED to 22 segments** from local `artifacts/promo/intervals.csv` (reel untouched; junk 0-seg dirs removed). **? ROOT-CAUSE OF "no logs" FIXED ? engine [#271](https://github.com/Khelsutra/badminton-highlight-indexer/pull/271) MERGED** (isolated worktree, full gate 1027 green): `backend/worker/__main__.py` **never configured logging** ? all pipeline diagnostics (detector choice, "Running native WASB inference", availability failures) were silently dropped at default WARNING. Now: `_setup_logging()` at the CLI entry (INFO; `force=True` so it owns the level) + a `-v/--verbose` DEBUG flag (native cmd, frame counts); and `cmd_infer` emits a **LOUD WARNING on a 0-segment success** (video_id + segmenter + detector_impl + likely causes + "re-run -v"). So **next cold-start, a 0-rally run is fully analysable from the log** ? that's the diagnostic that was missing this time.

4. user (2026-06-21T13:34:57.781Z)

ANNOTATE_AND_TRAIN.md
DRY_RUN_CUJ.md
FRIENDLY_AND_MULTILINGUAL.md
LOCAL_APPLIANCE_ARCHITECTURE.md
PER_RALLY_SELECTION.md
PORTABILITY.md
README.md
REQUEST_ACCESS_FORM.md
---MEMORY---
---ALT---
c--Users-avidu-Projects-Verifiers-khelsutra-badminton-highlight-indexer-badminton-highlight-indexer

5. assistant (2026-06-21T13:35:01.114Z)

Let me read the key CUJ documents and the engine's source surfaces.

6. user (2026-06-21T13:35:01.634Z)

Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local (multilingual)

> **Status:** `DRAFT` ? owner-gated · **Owner:** Avi Dullu · **Created:** 2026-06-21 · **Goal for this session.**
> **Umbrella:** a milestone of `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the appliance) + `docs/PER_RALLY_SELECTION.md` (the shipped SPA back-half).
> **Honesty note:** grounded in a file:line integration-surface map (this session); claims tagged `[verified]` / `[gap]`.

0. The goal (owner's words)
Operate the **web UI** ? point at a **video on the GCS bucket** ? the engine (tool) processes it ? **review/select rallies in the UI** ? **compile** ? **download the reel to the local machine**. Then **repeat with the UI in Hindi** to prove the multilingual workflow.

1. What already works vs the gap `[verified]`
? Back half (shipped ? PER_RALLY_SELECTION M0): the SPA loads `?video=<id>` ? `POST /api/query` (rallies, default all-selected) ? deterministic query/select ? `POST /api/compile` ? `<video>` preview + **`<a download>` saves the MP4 locally**. Reel download-to-local **works today** (`GET /api/highlights/{video_id}/{filename}` streams it; `/api/compile` also returns `download_url`). Fully **en/kn/hi** (the switcher / `?lang=`).

? Front half ? the gap is the SPA, NOT the engine `[corrected 2026-06-21; see §8]`:
- The SPA has **no ingest/process UI** ? it can't point at a video (bucket or local), can't call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in the URL. **This SPA front-half is the *only* net-new work for Path B (= B-P2).**
- **HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified file:line]`:** `POST /api/process` accepts a `gs://`/`gcs://` (also `s3://`/`local://`) `video_path` directly. Submit resolves the URI and **skips the local-existence check for non-local backends** (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`?`gcs` normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch (`_execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the **same** hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into the API's SQLite `videos`/`play_segments`. Proven end-to-end by `tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202, not 404 ? job `done` ? `db.get_video(video_id)` row with the **source URI preserved**), plus the hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
  - *Where the earlier "net-new" claim went wrong:* it conflated the **API path** (`/api/process`, which DID gain URI ingest in M2) with the separate **worker CLI** (`python -m backend.worker infer --video-uri gcs://?`), which writes `intervals.csv`/`report.json` to the bucket ? **not** the API DB. The CLI distinction is real; the "the endpoint is local-only" conclusion was not. ? **Engine-surface claims drift between sessions** (the engine ships PRs independently) ? re-verify any `[gap]`/`net-new` engine claim against the cited `file:line` before scoping it as work.

2. Two paths to the goal
Path A ? Cheapest dry run NOW (~zero new code). Pre-process the bucket video out-of-band to mint a `video_id` in the engine DB, then drive the existing SPA back-half. Proves the **multilingual UI-driven workflow** end-to-end. The only non-UI step is "point at the bucket" (a one-line CLI pre-process).

Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first scoped). The **HTTP bucket-ingest endpoint already exists** (§1) ? so Path B is **only the SPA front-half** (appliance S1/S2): an **ingest** input (enter a `gs://` URI) ? `POST /api/process` ? an **ingest + progress** screen polling `GET /api/jobs/{id}` ? on `done`, load `result.video_id` (which the existing back-half already renders). **No net-new engine endpoint** ? B-P1 is already done; the remaining work is **B-P2 (the SPA)**.

3. Path A runbook (the cheapest dry run) `[verified-steps]`
1. **Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`, binds `127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only, no WSL** ? local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)
2. **Get the bucket video local + into the API DB:**
   `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST /api/process {"video_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET

/api/jobs/{job_id}` until `result.video_id`. *(`gemini` needs `GEMINI_API_KEY`; `motion` is key-free but hides fabricated metadata.)*
27 3. **Open the SPA against the engine:** `npm run dev` (Vite proxies `/api` ? `:8000`, same-origin) ? `http://localhost:5173/?video=<video_id>` ? rallies auto-load.
28 4. **Select ? Build ? download:** pick rallies ? Build ? `<video>` preview ? click the download link ? MP4 saved local.
29 5. **Repeat in Hindi:** click **??????** (or `?lang=hi`) ? redo select?compile?download ? confirm the whole flow works with the UI in Hindi.
30
31 **Small unblocked SPA polish that makes Path A more "via the UI"** (independent of A/B, useful in both): a **"Load a video by ID"** input (paste the `video_id` ? load, instead of editing the URL) + consume the engine's `download_url` (instead of hand-building it). One tight PR.
32
33 ## 4. Open decisions ? OWNER (these gate the dry run)
34 1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-only**; the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7), then B (B-P2, the SPA ingest/progress screen).**
35 2. **Compute / detector:** **Gemini** (real rallies; needs your `GEMINI_API_KEY`; uploads downsized chunks to Google = paid + cloud) · **motion** (key-free, runs anywhere, but lower-quality boundaries + fabricated server/shot metadata stays hidden) · **GPU WASB** (best quality, but needs a GPU host + is owner-parity-gated, not production-verified). *Recommend Gemini for "real-ish, cheapest" ? pending your key + OK (paid + uploads footage).*
36 3. **Engine host:** local Windows (simplest, Gemini/motion) · the CPU droplet `168.144.126.176` (shared testbed ? careful) · a GPU VM (only for WASB).
37 4. **Which bucket video:** e.g. `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` (already staged, known-good) or another clean ?720p clip (must pass the engine validators).
38
39 ## 5. Honest caveats
40 - **Gemini = paid + uploads footage to Google** (an external/paid action ? needs your explicit OK + a clean clip). **Rotate the chat-shared `GEMINI_API_KEY`.**
41 - **motion** fabricates `server`/`receiver`/`shot_count` (the SPA correctly hides them) ? boundaries are real but metadata isn't.
42 - **GPU/native WASB** is partially wired + owner-byte-parity-gated ? **not production-verified**.
43 - Path A's "point at the bucket" is a **CLI pre-step**, not in-UI ? full in-UI ingest is Path B.
44 - Edge auth: none in the engine (localhost dry-run is fine; required before any non-localhost exposure).
45
46 ## 6. Definition of done (the dry run)
47 - [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ? compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
48 - [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ? ? (Kannada also works via the same switcher.)
49 - [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI pre-step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ? `POST /api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded (1080p, 76s)**, all from the UI. Surfaced + fixed an engine compile bug along the way (engine [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)). **Screencast:** `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
50 - [ ] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ? droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a from-scratch engine install or a proper deploy of the fixed engine.
51
52 ## 7. ? Run result ? Path A, local (2026-06-21)
53 Ran the real CUJ end to end on the local Windows box, isolated engine off `origin/master` (fresh DB/output worktree), **Gemini** detector:
54 1. `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` (1080p/30fps) ? `gsutil cp` local ? `POST /api/process {segmenter: gemini}`.
55 2. Engine processed via Gemini (3 chunks) ? **12 rallies** (16 found, 4 too-short rejected), `video_id 05e0e577?`.

56 3. SPA (`npm run dev`, Vite proxies `/api`?`:8000`) opened at `?video=05e0e577?` ?
loaded the 12 real rallies ("Loaded 12 rallies from this session").
57 4. **Build ? `/api/compile` ? reel** (1920×1080, 76.5s) ? **downloaded to
`~/Downloads/khelsutra-dryrun/highlights_local_en.mp4`** (50.9 MB).
58 5. **Repeated with the UI in Hindi** (`?lang=hi`): the page + rally cards + "??? ?????"
+ "???? ??? ????? ??" all in Hindi; reel compiled + **downloaded** (`highlights_local_hi.mp4`).
? multilingual workflow confirmed.
59
60 Screencast/proof: per-locale screenshots; both reels on the local machine. Screencast of
the i18n SPA is at `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`.
61
62 ### ? Fix the dry run surfaced (engine) ? MERGED-PENDING
63 The async-job server **hung every job in `queued`** when run the documented way (`python
-m backend.main`): main.py runs as `__main__` (sets `__main__.db_instance`), but `job_worker`
reads `import backend.main` ? a **separate** module whose `db_instance` is `None` ? the drain
crashes silently. Latent because the unit tests monkeypatch the executor inline and the UI was
verified vs a stub engine. **Fix + a real-subprocess regression test ? engine
[#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281)** (dispatch through
`backend.main`; FAILS-without / PASSES-with). Unblocked the run via the single-module
invocation; the fix applies to the droplet too.
64
65 ## 8. ? Correction (2026-06-21): B-P1 (the bucket-ingest endpoint) was already shipped ?
verified against code
66 **What this doc claimed (§1, original):** "`POST /api/process` accepts only a **local**
`video_path`" and "bridging `gcs://` ? a queryable `video_id` in the API DB is **net-new**
(T12/T14)."
67
68 **What the code actually does `[verified file:line, this session]`:** `POST
/api/process` has accepted bucket/Drive URIs since **engine M2
([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280))**. A
`gs://`/`gcs://`/`s3://`/`local://` `video_path`:
69 1. is resolved at submit (`backend/main.py:274` ? `storage.resolve_input_ref`;
`gs://`?`gcs` at `backend/storage/ref.py:60`; explicit scheme wins under default
`input_backend=local` at `ref.py:87`),
70 2. **skips the local-`os.path.exists` gate** for non-local backends (`main.py:283`) ?
returns **202, not 404**,
71 3. is fetched to scratch by the worker (`_execute_processing` ?
`storage.acquire_local_input`, `main.py:159` ? `GCSStorage.fetch_to_local`), then
72 4. runs the **same** pipeline as a local file, writing a queryable `video_id` to the API
SQLite DB.
73
74 **? Scope pin (default single-user mode):** this holds with
`security.allowed_video_root` **unset** (the local-appliance default). In **HOSTED mode**
(`allowed_video_root` set), a bucket URI is **correctly rejected `400`** ? a tenant can't point
the engine at an arbitrary bucket. That is by design, not a regression.
75
76 **Already tested** (so this can't silently regress):
`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches`,
`?::test_process_hosted_mode_rejects_nonlocal_uri` (the hosted-mode counterweight),
`?::test_process_unknown_scheme_returns_400`,
`?::test_cli_nonlocal_input_fetches_and_keeps_uri`. So **B-P1 needs zero new engine feature
code** ? the only Path-B work is **B-P2 (the SPA ingest/progress screen)**. *(The hermetic test
fakes the network download via `m.storage.fetch`, so it proves the CODE routing, not the live
GCS fetch ? that needs `google-cloud-storage` + ADC on the box, verified at deploy time / option
3.)*
77
78 **Why it went stale (anti-spiral lesson):** the §1 claim was tagged `[verified]` but was
inferred from the **worker CLI** (`backend.worker infer`, which writes CSV to the bucket) rather
than checked against the **API endpoint's** routing code ? and M2/#280 had since wired URI
ingest into `/api/process`. **The engine ships PRs between sessions, so its surface drifts.**
Convention going forward (also baked into the memory steering): **before scoping any
`[gap]`/`net-new` engine claim as work, re-verify it against engine code at a cited `file:line`
? a `[verified]` tag without a `file:line` is a hypothesis, not a fact.**
79
80 ## 9. ? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)

81 The full Path-B flow, driven **entirely from the UI** (no CLI pre-step), against a real local engine (`python -m backend.main --server`, Gemini detector, `GOOGLE_APPLICATION_CREDENTIALS` for the GCS fetch):
82 1. Opened the SPA ? the **"Process a video"** ingest panel (B-P2) ? pasted `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` ? **Process video**.
83 2. The engine **fetched the bucket video** (`google-cloud-storage` + ADC) ? **Gemini** segmented it (live "Processing? segmenting (gemini)" progress in the UI) ? **12 rallies** (`video_id 05e0e577?`), handed off via `?video=` ? the existing back-half rendered them.
84 3. **Build reel** ? `/api/compile` ? **reel preview + download** ? MP4 saved locally (**1920×1080, 76s**).
85 4. **Screencast of the whole CUJ** (Playwright-recorded; the ~3-min Gemini wait fast-forwarded): `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
86
87 ### ? Bug the run surfaced + fixed (engine) ? MERGED
88 `/api/compile` **500'd** on a bucket-ingested video: `videos.filepath` keeps the **source URI** (`gs://?`, M2 provenance), and the stitcher was handed that directly ? ffmpeg can't open a `gs://` path. **Fix:** compile now resolves the source via `storage.acquire_local_input` first (identity for a local path; fetch-to-scratch for a bucket URI) + a regression test ? engine **[#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)** (full suite 1182?). This is what makes the in-UI bucket?reel CUJ actually complete. *Proof-and-validation-driven: recording the screencast is what found the bug.*
89
90 ### Changelog
91 - `2026-06-21` ? created; Path-A run recorded (§7).
92 - `2026-06-21` ? **§8 correction:** verified B-P1 (`gs://`?`/api/process`?API-DB `video_id`) is already shipped + tested (M2/#280); §1/§2/§4/§6 reframed so Path B = SPA-only (B-P2). Sibling stale claims fixed in `LOCAL_APPLIANCE_ARCHITECTURE.md` (`/api/health` exists; bucket-URI ingest not a net-new endpoint; T12 engine-half done) + `README.md`.
93 - `2026-06-21` ? **§9: Path-B live e2e DONE** ? the in-UI `gs://`?reel CUJ ran end to end (12 rallies ? reel downloaded); surfaced + fixed the compile-from-bucket 500 (engine #289); screencast uploaded to the promo bucket. **DoD §6 box 3 ?.**
94

7. user (2026-06-21T13:35:02.331Z)

1 # Interactive per-rally selection ? reel builder (RallyPicker)
2
3 > **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ? select ? compile ? download, on existing engine endpoints; verified e2e against a stub engine, GCP-GPU validation next). Design **APPROVED by owner 2026-06-20**. Downstream milestones M1?M5 (multi-video auth, per-rally preview endpoint, corpus loop, localized NL) remain owner-gated. · **Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
5 > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6 > **Relation:** second major tracked work item in the `khelsutra` repo (peer of `REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline. Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan **M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` + `/api/compile` contract.
7 > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not legal/financial advice.
8
9 ---
10
11 ## 0. TL;DR
12 Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally **list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest first") pre-selects into the **same** selection set; a sticky footer compiles the selection into one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ? `POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396, 307-328]` ? with **zero new engine intelligence and zero new endpoints**. The single most

important caveat: queries and chips are scoped to **length / count / order / shot-count only**; shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the *final compiled reel* can stream ? so MVP selects on text metadata, and visual preview/timeline is a deliberate later milestone.

1. Problem & goal

Today the alpha can detect rallies and stitch a reel, but reel-building is either a **silent `stitching.min_shot_count` threshold** `[verified backend/main.py:362-383]` or a static checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies` while the API returns `play_segments`, so it renders nothing `[verified backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is **dynamic per-video selection, not a fixed threshold**: a coach should say "give me the long rallies from this session" and get exactly those, then fine-tune by hand.

The concrete outcome ("good"): A coach uploads a 40-minute session; the engine finds 60 rallies. They type *"top 15 longest"*, eyeball the auto-selected cards, untick two warm-up rallies, name it *"Tuesday ? long points"*, and download a ~6-minute reel ? without learning any threshold knob.

Goal: the smallest *honest* surface that turns engine-detected rallies into a user-curated, downloadable reel, with both **fast** (query) and **precise** (checkbox) selection sharing **one** state.

Non-goals (alpha): no conversational LLM (`UI_PROPOSAL.md:66` defers it); no shot-type/player/score filters (no data exists); no per-rally video preview while it requires unbuilt streaming; no multi-tenant auth (single-box alpha).

2. Decisions locked

#	Decision	Source / date	Implication
D1	Dominant surface = selectable rally **list/grid**	Synthesis 2026-06-20 (Grid scored 15, fit-to-data 4)	Scales to long matches via bulk+filter; shows honest fields only
D2	Fast entry = **deterministic** client-side `parseQuery()` (no LLM)	`[verified UI_PROPOSAL.md:66]` conversational QA deferred	Predictable, unit-testable, offline, zero API cost
D3	**One** shared `Set<segment_id>`, mutated by query + checkbox + (future) timeline	Synthesis 2026-06-20	No second compile path can diverge from what the user sees
D4	Compile via existing `POST /api/compile {video_id, segment_ids, output_filename}`	`[verified backend/main.py:342-396]`	Selection order is harmless ? stitcher `_merge_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]`
D5	Query grammar (alpha) = length, count, order, **shot_count only**	`[verified database.py:122-135, gemini.py:470-485]`	Only fields the payload actually returns; grey out the rest
D6	MVP backend = **zero new endpoints**	`[verified]` reuse map (§4)	Ships the full loop on the already-shipped contract
D7	Replace the silent `min_shot_count` cut with **explicit user selection**	Owner framing + `[verified main.py:362-383]`	UI *warns* about the floor; it never hides why a rally vanished
D8	**Default selection = ON** (curate-by-removal)	Owner 2026-06-20	Rallies load all-selected (matches today's "all rallies" reel); user removes warm-ups. Footer/empty UX assumes a non-empty start
D9	**Preview-before-select is NOT a ship-blocker** for the MVP	Owner 2026-06-20	MVP selects on text metadata + previews the *final compiled reel*; the source/clip-stream endpoint (per-rally preview) is the next milestone (P9), not MVP
D10	**MVP is job-tethered** (carry `result.video_id`); no `GET /api/videos`	Owner 2026-06-20	Smallest honest slice; multi-video home + identity scoping is P8 (gated)
D11	`min_shot_count`: **warn, do NOT override** in the MVP	Owner 2026-06-20	The UI surfaces the floor; an explicit-selection override of the compile default is deferred (would trigger the Launch Report convention)

3. Foundation ? what the engine gives us today `[verified this session]`

The whole design rests on the engine's real, already-shipped contract. Read against the engine repo (`badminton-highlight-indexer`):

- **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time, duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration computed at insert).
- **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the *only* rally-list path today; it JSON-parses `metadata` and joins `events` `[verified main.py:331-340, database.py:427-475]`.
- **Building a reel = `POST /api/compile {video_id, segment_ids:[int], output_filename}`** ? loads `videos.filepath`, filters all segments to the requested `segment_ids`, applies the `min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and calls `VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396, compiler.py:303]`, which **merges overlaps** then per-clip re-encodes ? concat ? one MP4 `[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.
- **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled MP4. Compile returns a **server-local filepath, NOT a URL** `[verified main.py:394]`, so the SPA constructs this download URL itself. The compiled reel is the **only streamable video that exists today** `[verified main.py:307-328]`.
- **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count` `[verified main.py:121-123]`, so the UI can warn before a selected rally is silently dropped.
- **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified main.py:274-293]`, which is how the MVP reaches a video without a (nonexistent) list-videos endpoint.

Honest fields vs fabricated/roadmap (critical for the UI):
- **Honest today** (default `default_segmenter: "gemini"`, per `config.json` `[verified]`): `start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's `shuttle_exchange_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot_count_confidence` (a **string**, often `"Unknown"`) `[verified gemini.py:470-485]`. On the gemini path `server`/`receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.
- **Fabricated ? do NOT surface:** the `motion` demo segmenter fills `server`/`receiver` via `random.sample(...)`, a random `shot_count`, and random `events` (serve/smash/foul/winner) `[verified motion.py:191-198]`. `query` returns these columns + `events`, but they are not ground truth and must stay hidden.
- **Roadmap ? no column, no detector:** shot-type per shot, player identity, match/point score, game/set grouping. None exist anywhere in the schema; do not consume them `[verified database.py whole-schema 122-148]`.
- **Correction-loop columns exist but are dead:** `candidate_segments.human_label/notes/confirmed_segment_id` are defined but no endpoint writes them `[verified database.py:166-168]` ? that's the M4 corpus extension (engine PR-4/B6, `UI_ALPHA_EXECUTION_PLAN.md`).

4. Design / architecture

Reuse map (existing ? `[verified]`, no change for MVP): `POST /api/query` (list) ? `POST /api/compile` (stitch) ? `GET /api/highlights/{video_id}/{filename}` (download); `GET /api/config` (floor) ; `GET /api/jobs/{id}` (carry `result.video_id`). **All net-new code is the SPA** in `khelsutra` `web/` (Vite + React + TS + Tailwind, per `web/README.md`), plus a small set of *optional* real-gap engine enablers (§4.3) ? none required for MVP.

4.1 Components (net-new SPA)
- **QueryBar + `parseQuery()`** ? a *pure, deterministic, unit-tested* string ? `{filter, sort}` compiler over `duration` / `ordinal` / `shot_count`. Renders a **parsed-pill** ("filter: duration > 10s · sort: duration desc") so the interpretation is transparent and hand-correctable; greys out shot-type/player/score tokens with an inline "not available yet". No LLM, no network call.
- **RallyList/Grid** ? one card per `play_segment`: ordinal (by `start_time`), `mm:ss` start, duration (`end?start`), and a shot-count + confidence chip **only when `metadata.shot_count` is present** (absent on detector paths that don't write it). Checkbox per card; the thumbnail area is a placeholder (no endpoint yet).
- **`useRallySelection()` hook** ? owns `selectedIds:Set<number>`, `displayOrder`, `apply(queryResult)`, `toggle(id)`, and bulk Select-all / Clear / Invert. Unit-tested (matches the exec-plan's "API client + review-state logic" bar).
- **SelectionFooter (sticky)** ? live "N clips / total M:SS"; a **`min_shot_count` warning** when a selected rally would be dropped at compile (read from `GET /api/config`); reel-name input; one **Build** button.

63 - **CompileResult** ? `<video src=GET /api/highlights/...>` preview of the *compiled reel* + a download link.
64 - **TypedApiClient** ? `query`, `compile`, `buildDownloadUrl`; the typed seam to the engine.
65
66 **4.2 Data flow** `[verified anchors inline]`
67 Gemini segmenter (process time) writes `play_segments {id, start/end_time, duration, metadata.shot_count/confidence/detector}` `[gemini.py:461-498]`
68 ? SPA `POST /api/query {video_id}` ? `{play_segments:[...]}` `[main.py:331-340]` (**fix** the static UI's `data.rallies` read ? `play_segments` `[app.js:344]`)
69 ? normalize to `RallyCard[]` + one shared `selectedIds:Set`; `server`/`receiver`/`events` are present in the payload but motion-only fabrications ? **not surfaced** `[motion.py:191-198]`
70 ? all input modes mutate the **same** `selectedIds`: (a) `parseQuery()` predicate over duration/ordinal/shot_count; (b) checkbox toggle + bulk ops; (c) *(post-MVP)* timeline band
71 ? footer recomputes count + ?duration **client-side**, and warns for any selected rally with `shot_count < stitching.min_shot_count` `[main.py:362-383]`
72 ? **Build** ? `POST /api/compile {video_id, segment_ids: [...selectedIds], output_filename}` ? stitcher de-dupes via `_merge_overlapping` `[compiler.py:311-317]`, cuts `time_pairs` from `videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393, compiler.py:303]`
73 ? compile returns a **server-local filepath, not a URL** `[main.py:394]` ? SPA builds `GET /api/highlights/{video_id}/{output_filename}` to preview in `<video>` + download `[main.py:307-328]`.
74
75 **4.3 Engine API additions (none for MVP; tagged against real gaps)**
76 | Endpoint | Purpose | Status |
77 |---|---|---|
78 | `POST /api/query` | List rallies (only path) ? fix SPA to read `play_segments` | **exists** |
79 | `POST /api/compile` | Stitch selected `segment_ids` ? MP4 | **exists** |
80 | `GET /api/highlights/{video_id}/{filename}` | Stream/download compiled reel | **exists** |
81 | `GET /api/config` | Read `stitching.min_shot_count` for the warning | **exists** |
82 | `GET /api/jobs/{job_id}` | Carry `result.video_id` into the picker | **exists** |
83 | `POST /api/compile` ? add `download_url` | Return `/api/highlights/...` so SPA stops hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **modify** |
84 | `GET /api/health` | Engine-offline banner ping the SPA is specified to use (`HOSTING_PLAN.md:42`) | **new** |
85 | `GET /api/videos/{video_id}/rallies` | RESTful/cacheable alias returning the same `{play_segments}` shape | **new** |
86 | `GET /api/videos` | List a user's videos for a Match picker (only `get_video(id)` exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
87 | `GET /api/videos/{video_id}/source` (Range) *or* `/clip?start=&end=` | Stream source/per-span video so a rally previews before selection ? neither exists | **new** |
88 | `GET /api/videos/{video_id}/rallies/{segment_id}/thumbnail` | Per-rally poster still (no thumbnail column; WASB frames are transient) | **new** |
89 | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ? `candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90
91 **? Dead-ends / do-NOT:** never surface `server`/`receiver`/`events` (motion-only randoms `[motion.py:191-198]`); never consume shot-type/player/score (no data); never render a confidence **meter** ? `shot_count_confidence` is a Gemini *string* (often `"Unknown"`) and `play_segments` has **no** numeric confidence column.
92
93 **4.4 Observability (first-class, per `[memory: feedback-launch-quality-bar]`).** Every P3 user action emits a **structured JSON log line** (`web/src/lib/log.ts`) so a journey is traceable from text to selection ? the SPA-side "runlog". Each query application generates **one `cid`** shared by every line of that journey; the future `/api/compile` call (P5) reuses the same `cid` to thread the trace into the engine. Events (all on the P3 path):
94
95 | event | level | when | key fields |
96 |---|---|---|---|
97 | `query.parse` | info | every apply attempt | `cid`, `source` (typed/example),

`recognized`, `filters`, `sort`, `limit`, `unavailable` ? the exact interpretation |
98 | `query.unavailable_request` | info | the query names a shot-type/player/score | `cid`, `raw`, `unavailable[]` ? **demand signal** for roadmap features (what users ask for that we can't yet do) |
99 | `query.skip` | info | nothing actionable parsed | `cid`, `raw`, `reason` |
100 | `query.apply` | info | a recognized query is applied | `cid`, counts (`total`/`selected`), `selected_ids`, `dropped_unavailable` |
101 | `query.empty_result` | **warn** | recognized query matches **zero** rallies | `cid` + full interpretation ? a looks-empty grid is **never silent** |
102 | `query.apply_failed` | **error** | the (pure) parser/selector unexpectedly throws | `cid`, `raw`, `error` ? loud, never swallowed |
103 | `selection.bulk` | info | Select-all / Clear / Invert | `cid`, `op`, `result_count` |
104 | `rally.toggle` | debug | a card is added/removed | `id`, `action` |
105
106 No silent drops: unsupported concepts (`query.unavailable_request` + `dropped_unavailable`) and empty results (`query.empty_result` warn) are always logged. Verified live in-browser (events fire with correct `cid`/fields, zero console errors).
107
108 ## 5. Risk table
109 | Risk | Severity | Mitigation |
110 |---|---|---|
111 | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream `[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel; preview/timeline deferred to M3 behind a new endpoint |
112 | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate (Launch-Report-worthy `[memory: launch-report-convention]`) |
113 | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true scrubber until source streams |
114 | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))` fallback) `[gemini.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not a meter |
115 | No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate `GET /api/videos` behind PR-3 scoping |
116 | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for MVP; `download_url` field (M1) removes the brittleness |
117 | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate IA: query+grid = one screen; correction = separate ? corpus screen (owner call, §8) |
118
119 ## 6. Honest limits ? what this does NOT do
120 A green SPA build + passing `parseQuery`/selection unit tests prove the **picker logic only** ? **not** that the engine's rally boundaries are correct (the measured **rally-END** weakness is a separate quality track, unaffected by this UI). "Conversational" here means a **deterministic four-dimension parser**, not natural-language understanding. Nothing is implemented in `web/` yet ? this is `[design]`. The feature curates *what the engine already found*; it cannot surface a rally the detector missed, nor split/merge a mis-bounded one (that's the M4 correction loop).
121
122 ## 7. Deliverables & progress tracker ? **source of truth**
123 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. CI (`npm test`) must stay green; merge to `master` auto-deploys the site (docs/`web/` changes don't touch the live `site/` Worker).
124
125 | ID | Deliverable | Depends on | Gated? | Status | PR |
126 |----|-------------|------------|--------|--------|----|
127 | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? | [#11](https://github.com/avidullu/khelsutra/pull/11) |
128 | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to engine `:8000` | P0 | No | ? | [#14](https://github.com/avidullu/khelsutra/pull/14) |
129 | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) | P1 | No | ? | [#21](https://github.com/avidullu/khelsutra/pull/21) |
130 | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable tokens) | P2 | No | ? | (this PR) |

P4	SelectionFooter + `min_shot_count` warning (from `GET /api/config`)	P2	No	?	(this PR)
P5	Build ? `POST /api/compile` ? client-built download URL + `<video>` preview of the reel	P2	No	?	(this PR)
P6	*(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns `download_url`	?	No	?	?
P7	*(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias	?	No	?	?
P8	*(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen	P7	? owner	?	?
P9	*(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail	P8	? owner	?	?
P10	*(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally Review	P9	? owner	?	?

Milestone rollup: **M0/MVP = P1?P5 ? COMPLETE (2026-06-20)** (pick?reel on existing API, zero new engine endpoints) · **M1 = P6?P7** (RESTful + reliability polish) · **M2 = P8** (multi-video home) · **M3 = P9** (per-rally preview/timeline) · **M4 = P10** (corpus correction loop) · **M5** = swap `parseQuery()` for localized NL behind the same seam (gated on shot/player data landing).

8. Open questions ? owner / external
Blocking gates ? ? RESOLVED 2026-06-20 (owner) ? see D8?D11 in §2:
1. ~~Default selection ON vs OFF?~~ ? **ON / curate-by-removal** (D8).
2. ~~Is preview-before-select a ship-blocker?~~ ? **No**; text-metadata MVP, preview is P9 (D9).
3. ~~Job-tethered vs multi-video MVP?~~ ? **Job-tethered** (D10).
4. ~~`min_shot_count` override?~~ ? **Warn, don't override** in MVP (D11).

Still open (nice-to-knows, non-blocking):
5. May user-facing copy call `parseQuery()` "conversational", given conversational QA is explicitly deferred (`UI_PROPOSAL.md:66`)? Risk of over-promising NL. *Owner positioning/honesty call.* *(Lean: avoid the word; show the parsed-pill instead.)*
6. One screen for query+grid (recommended) vs a separate ? corpus/correction screen for A4? *Owner positioning call.*

9. Definition of done
- [ ] From a finished job, a user can list rallies, query/select, and download a reel ? on **existing** endpoints.
- [ ] Only honest fields (ordinal / duration / shot_count+confidence-when-present) are shown; fabricated `server`/`receiver`/`events` are hidden.
- [x] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens, shows a visible parsed-pill, and is unit-tested. *(P3)*
- [x] The P3 path is **observable**: structured, correlation-id'd logs for parse / apply / skip / empty / unavailable-request / manual-edit; unsupported concepts and zero-result queries are logged loudly, never dropped silently (§4.4). *(P3)*
- [x] Selection hook is unit-tested; the `min_shot_count` floor is surfaced, not hidden. *(P2 hook · P4 floor warning ? `lib/floor.ts` mirrors `main.py:362-383`, exempts no-shot_count rallies, blocks Build when all-dropped)*
- [x] From a finished job, a user can list rallies, query/select, and download a reel ? on **existing** endpoints. *(P5 ? `?video=<id>` loads via `POST /api/query`; Build ? `POST /api/compile` ? `<video>` preview + download over `GET /api/highlights/...`; verified e2e against a stub engine)*
- [x] `npm run build` (and `npm test`) clean; the static UI's `data.rallies` bug is gone ? the SPA **supersedes** the static `app.js` for this flow (reads `play_segments`).
- [x] §7 tracker + this DoD updated. *(Doc stays live for the owner-gated M1?M5 engine milestones; not archived ? only the MVP/M0 slice is complete.)*

10. References
Engine code anchors `[verified this session]`: `backend/main.py:121-123` (config), `:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile + `min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135` (play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475` (query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485` (honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198` (fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe

+ entrypoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json` (`default_segmenter: "gemini"`).
164 **Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`; `HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now, model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.
165
166 ---
167 ### Changelog
168 - 2026-06-20 ? Initial `DRAFT`. Consolidated three candidate designs (Timeline / Grid / RallyPicker) into the hybrid recommendation via a multi-agent design pass; every engine-fact anchor re-verified against source before writing.
169 - 2026-06-20 ? Owner **approved** the recommendations; locked D8?D11 (default ON · preview not a ship-blocker · job-tethered MVP · warn-don't-override the floor); status `DRAFT ? IN PROGRESS`; P0 done.
170 - 2026-06-20 ? **UI: lead with rally LENGTH, park shot-count as experimental** (owner). Rally cards emphasize duration (`m:ss long`); the shot-count chip is flagged amber "experimental" with a tooltip ("under development, may be inaccurate ? filter by length instead"); example chips are duration-led (dropped "at least 8 shots"); a note states "rally length is fully supported, shot-count is experimental". `parseQuery` still accepts shot-count queries (not broken) but they're de-featured. Public badge `P5`?`alpha`. A test pins that every example chip parses to a recognized, fully-available (non-shot-count) query. Suite 75 green.
171 - 2026-06-20 ? **P5 shipped ? MVP / M0 COMPLETE.** `App` now loads a real session from `?video=<id>` (`POST /api/query`, default ALL-selected D8) with a mode banner (demo / loading / loaded N / error); **Build** ? `compileReel` (`POST /api/compile`) ? `CompileResult` previews the stitched reel in `<video>` over `GET /api/highlights/{video_id}/{filename}` (client-built URL ? engine returns a filepath, not a URL) + a download link; `reelFilename()` slugifies to an engine-safe bare `.mp4`. Loading/building/done/error states, all traced (`rallies.load_*`, `build.start/done/failed`). Verified e2e against a stub engine: load 5 rallies ? real `/api/config` floor warning ? Build ? `<video>` reached `readyState=4` (decoded a real mp4 through the proxy) + download link correct; all-below-floor selection keeps Build disabled (P4 guard prevents the 400). `reelFilename` unit-tested; suite **73 green**, typecheck + build clean. **Next: GCP-GPU end-to-end validation (real video ? rallies ? reels).**
172 - 2026-06-20 ? **P4 shipped**: `SelectionFooter` (live summary + reel-name input + Build) with an **honest `min_shot_count` warning** ? `lib/floor.ts` mirrors the engine floor (`main.py:362-383`) EXACTLY: a rally is dropped only if it has a numeric `shot_count` below the floor; rallies *without* `shot_count` are exempt; **all-below ? Build disabled** (the engine would 400). Config read from `GET /api/config` on mount (graceful offline). 8 floor unit tests; amber/red warning states verified live in-browser; suite **70 green**, typecheck + build clean.
173 - 2026-06-20 ? **P3 shipped**: deterministic `parseQuery()` (`web/src/lib/query.ts`) over length/count/order/shot_count + `runQuery`/`sortRallies`/label helpers and the `QueryBar` (live parsed-pill, example chips, greyed "not available yet" for shot-type/player/score ? never faked). Writes the shared selection set via a new `useRallySelection().apply`; `RallyGrid` ordinals made stable so a re-sort doesn't renumber cards. **First-class observability (§4.4):** structured, correlation-id'd `query.*` / `selection.*` / `rally.toggle` runlog (loud `empty_result`/`apply_failed`; `unavailable_request` demand signal). **Adversarial multi-agent review** (grammar/honesty/robustness/reuse) ran pre-merge ? 13 confirmed findings folded in, incl. 4 majors: an honesty leak (`at least 8 smashes` ? fake `length ? 8s`), a selection-wiping `top 0`, a unit-blind `first 30 seconds`, and self-contradictory pill labels (`first 10` shown as "top 10"). Live browser CUJ replayed (filter / top-N / honest-greying / untick / no-op / wipe-closed). Suite **62 green**, typecheck + build clean.
174

8. assistant (2026-06-21T13:35:05.937Z)

Now let me read the appliance architecture, the friendly/multilingual doc, and the annotate-and-train doc.

9. user (2026-06-21T13:35:07.398Z)

1 # Single-box appliance ? UI-powered rally indexer + reel generator (KhelSutra Studio)
2
3 > **Status:** `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge) · **Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)

> **Tracking anchors:** §9 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
> **Relation:** peer-of `PER_RALLY_SELECTION.md` / `REQUEST_ACCESS_FORM.md` in this repo; **consumes** the engine's decoupled `StorageBackend` + `python -m backend.worker` seams and the FastAPI loop; **extends** the engine's `HOSTING_PLAN.md` gating + `UI_PROPOSAL.md` alpha screens. Engine-side changes here are **proposals to coordinate** with the active decoupled/worker sessions ? not written unilaterally.
> **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine source this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not legal/financial advice.

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0. TL;DR
KhelSutra Studio turns the engine from a CLI into a **single-box, UI-powered rally indexer + reel generator**: the khelsutra `web/` React SPA, served behind an **edge auth gate**, drives the **existing** engine FastAPI loop (ingest ? index ? pick rallies ? reel ? download) on **one machine**, with `StorageBackend=local` for the MVP. It is the **single-box deployment profile of the just-shipped decoupled architecture** ? storage=local + compute=local is the degenerate collapse of the same storage/worker split.

The **single most important caveat:** the engine has **zero app-layer auth and CORS `*`** `[verified backend/main.py:42]`, so safety is structural **only** because the engine stays bound to `127.0.0.1:8000` `[verified backend/main.py:638]`, a firewall blocks inbound `8000`, and a reverse proxy/tunnel is the sole authenticated ingress ? **the gate, not the app, is the tenancy boundary.**

Two honest hardware truths: (1) the current droplet is **CPU-only** ? real detection there is **Gemini (cloud)** or **stub (CPU mock)**; **real local-GPU detection needs a GPU host** (native WASB, in-flight). (2) The owner priority is to **stand up a GPU host ASAP** ? provider-agnostic (DO GPU droplet / GCE GPU VM / Cloud Run+GPU) ? but that runs **in parallel** with the UI MVP, which ships on Gemini today.

1. Problem & goal
The decoupled engine just shipped a `StorageBackend` ABC + a `python -m backend.worker {infer|train}` dispatcher `[verified backend/storage/, backend/worker/__main__.py]`, and the FastAPI loop already covers `process ? jobs ? query ? compile ? highlights` `[verified backend/main.py]`. What's missing is a **console** that lets a non-CLI operator run the whole loop on a single box, **honestly, behind a gate**.

Headline CUJ (owner's words): a **hosted website** where I pick an input ? **a Google Drive file, a bucket file (`s3://`/`gs://`), or a local upload** ? it lands in the **bucket (DO Spaces / GCS)** or locally ? I **generate highlights** on a **GPU-enabled host** running the real e2e ? I **pick rallies, watch, and download** ? **completely from the UI.**

Outcome ("good"): an operator signs in, picks a source, watches it index, curates rallies, downloads a reel ? on the **live CPU droplet today** (Gemini/stub), and on a **GPU host later** (native WASB) with **no UI fork**.

Read to get current: `PER_RALLY_SELECTION.md` (§3/§4/§10 contract anchors), engine `HOSTING_PLAN.md` (gating), `DECOUPLED_COMPUTE/HANDOFF.md` (what's built), `VIDEO_LOCALITY_MODEL.md` (privacy/retention).

2. Decisions locked
#	Decision	Source / date	Implication
D1	Reverse-proxied **single origin**; engine stays `127.0.0.1:8000`	`[verified main.py:638]`; 2026-06-20	CORS `*` is moot from the browser ? the proxy is the boundary
D2	**Edge auth**, zero engine auth code	`HOSTING_PLAN.md`; 2026-06-20	Caddy basicauth OR cloudflared + Cloudflare Access
D3	MVP drives the **in-process FastAPI worker** loop, `storage=local`	`[verified job_worker.py:37, local.py:33]`; 2026-06-20	bucket worker = reserved scale path, out of MVP
D4	Embed **RallyPicker** from `PER_RALLY_SELECTION.md` **verbatim**	PR [#11]; 2026-06-20	the console is a superset; do not redesign the picker
D5	Console lives in the khelsutra **`web/`** React scaffold	`web/README.md`;	

2026-06-20 | Cloudflare-Pages-style root, not engine static |
D6	Compute placement is a **setting**, rendered from `GET /api/capabilities`	`[design]`; 2026-06-20	UI offers ONLY what the box can honestly do
D7	**Ingest is a source menu** (local upload · bucket URI · Drive) normalizing to one video object	`[verified backend/storage/ref.py parse_uri]`; 2026-06-20	Drive is a **stub** today ? scoped honestly (§4.2)
D8	GPU host is **provider-agnostic** (DO droplet / GCE VM / Cloud Run+GPU); storage stays the bucket	Owner 2026-06-20	compute and storage are separate planes; no provider lock-in
D9	**Observability is first-class** ? structured logs on every new path + a DoD item	Owner 2026-06-20	see §5; verified during CUJ replay
D10	Delivery = a **self-hosted webserver + UI the user spins up** (NOT a desktop app); the user points at **local storage + GPU/CPU OR cloud locations** and the UI drives the tool	Owner 2026-06-20	**supersedes `DESKTOP_APP_PLAN.md`**; one artifact (this appliance) runs on the owner box, a droplet, or a GPU host ? same UI, compute/storage are settings

3. Foundation ? what the engine gives us today `[verified this session unless noted]`

- **FastAPI loop:** `POST /api/process` (202 + `job_id`) ? poll `GET /api/jobs/{id}` ? `POST /api/query` ? `POST /api/compile` ? `GET /api/highlights/{video_id}/{filename}` (FileResponse) `[main.py]`. **Chunked upload exists** ? `POST /api/upload/init` · `PUT /api/upload/{id}/part/{i}` · `POST /api/upload/{id}/complete` · `DELETE /api/upload/{id}` `[verified uploads.py:64,89,123,147]` ? but the bundled `app.js` **never calls them** (it only hits `/api/process` + `/api/compile` `[verified app.js:88,373]`). So browser upload is **client-wiring only**, no engine change.
- **In-process worker:** one module-global `ThreadPoolExecutor(max_workers=1)` `[verified job_worker.py:37]` ? strictly **one job at a time**; the DB `jobs` table is the clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a **separate** knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1 to serialize) ? it is **not** already 1.
- **Storage:** `storage.backend` default = `"local"` `[verified models.py:298]`; `LocalStorage.fetch_to_local` is **identity** `[verified local.py:33]`, `put` no-ops on `src==dst` ? so on a single box the worker's pull?pipeline?push **collapses to read-local/write-local**. URI scheme `parse_uri` `[verified ref.py]`: bare/`local://` ? local; `s3://` (AWS/R2/**DO Spaces**/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? **follow-up stub (emulated only)**. Credentials never in config ? boto3 default chain / `~/.aws` `[verified s3.py]`.
- **Compute seam:** `indexing.detector_impl` ? {`auto`=WSL, `native`=Linux-native, `stub`=CPU mock}; default `"auto"` `[verified models.py:129]`, resolved in `trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB is in-flight on branch `feat/native-wasb-runner`:** `native_wasb_runner.py` **exists** `[verified]`; **`fusion_hybrid._build_native_runner` returns `NativeWasbRunner`** `[verified fusion_hybrid.py:86]` while **`trajectory_hybrid._build_native_runner` still raises** `[verified :57-63]`. Acceptance gate = **owner byte-parity** (WSL CSV == native CSV) on a real GPU box ? not exercisable on the droplet or in the offline suite. Treat native as **not production-verified**.
- **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ? pipeline ? pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified worker/__main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` + `--set indexing.skip_ai_handoff=true` + `storage.s3`.
- **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET /api/health` **now EXISTS** (engine M1, [#268]) ? `{status, sport, default_segmenter}` `[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]`.
- **Droplet reality:** `168.144.126.176`, **CPU-only, NO GPU**, region `blr1`; co-hosts the marketing site (`/srv/khelsutra-site`, node `:8787`) **and** the engine (`~/rally`); ran the full train+infer e2e against DO Spaces with the **GPU mocked** `[SETUP_NEW_MACHINE.md, DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run there.
- **Absent (net-new):** `GET /api/capabilities`, `GET /api/videos` (only single-row `get_video(id)`), `GET /api/reels/{video_id}`, any source/proxy streaming endpoint, any **Dockerfile**, any per-video **cost ledger** `[verified ? grep returns none]`.

4. Design / architecture

4.1 Topology & request path
ONE box: **edge gate** (cloudflared+Access or Caddy basicauth) ? **reverse proxy / static host** serving the `web/` SPA bundle and proxying `/api/*` ? engine **FastAPI `127.0.0.1:8000`** ? **LocalStorage** ? **detector seam** (Gemini/stub today; native WASB on a GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing site stays a **separate** systemd unit on a **separate** subdomain.

```
PUBLIC INTERNET
   ?  (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
   ?      cloudflared Tunnel + Cloudflare Access (no inbound ports)  OR  Caddy :443 + basicauth
   ?
????????????????????????????????????????????????????????????????????????????
? ONE BOX   firewall BLOCKS inbound 8000 + 9000/9001/9023                   ?
? (2) reverse proxy / static host (Caddy/cloudflared)                       ?
?      /, /assets/*  ? web/ React SPA      /api/*, /api/highlights/* ? :8000 ?
?            ?  ONE ORIGIN to the browser ? engine CORS '*' is irrelevant    ?
?            ?                                                              ?
? (3) ENGINE FastAPI  (host=127.0.0.1:8000, HARD-bound [main.py:638])      ?
?      control: jobs table + ThreadPoolExecutor(max_workers=1) drain        ?
?      data: uploads.py chunked upload · /api/compile · /api/highlights      ?
?            ?                                                              ?
? (4) STORAGE = LocalStorage (default)  fetch=identity · put=no-op(src==dst) ?
?            ?                                                              ?
? (5) COMPUTE seam: gemini (cloud, key) · stub (CPU mock) [droplet]          ?
?                   native WASB (GPU host) · auto (owner Win+WSL)           ?
????????????????????????????????????????????????????????????????????????????
RESERVED SCALE PATH (out of MVP): console "compute target" selector ? storage=s3 (DO Spaces)
   ? SEPARATE GPU host runs `python -m backend.worker infer --video-uri s3://?` ? outputs/<vid>/ back
   (needs a NET-NEW orchestration endpoint reflecting outputs into the jobs/poll contract)
```

4.2 Ingest sources (the headline) ? one menu, one normalized video object
All sources normalize to **"a video the pipeline can read"** ? a local path (MVP) or a `StorageRef` (scale):

Source	Path today	Status
Local upload	SPA chunks the file ? `max_part_mb` via the existing `/api/upload/init|part|complete` `[verified uploads.py]` ? server path ? `POST /api/process`. (Scale: also `StorageBackend.put` to `s3://`/`gcs://`.)	**wire client only** (endpoints exist)
Bucket URI (`s3://`, `gs://`)	**`POST /api/process` accepts the bucket URI directly** (resolve ? fetch to scratch ? same pipeline ? API DB) `[verified main.py:159,274,283 ? corrected 2026-06-21; was "net-new"]`; the worker CLI also fetches `--video-uri`. So the **engine endpoint is NOT net-new** ? only the SPA call that POSTs the URI + polls `/api/jobs` is. *(Default single-user mode; HOSTED mode w/ `allowed_video_root` correctly rejects a bucket URI 400.)*	**engine: DONE (verified); SPA ingest call: design (= B-P2)**
Google Drive (`gdrive://`)	`parse_uri` recognizes it, but the gdrive backend is an **emulated stub** `[verified ref.py, DECOUPLED_COMPUTE research]`	**stub ? roadmap;** MVP requires a Drive?bucket pre-step; do **not** present Drive as live until the backend is real

4.3 Compute placement matrix (one UI, surfaced from `/api/capabilities`)
Placement	Runs	Needs	Real/mock	Honest caveat
CPU droplet ? gemini (MVP target)	`segmenter=gemini`, in-proc worker	`GEMINI_API_KEY`; network; **no GPU**	**REAL**	Cloud brain: uploads ?1280px chunks to Google ? **must be in consent copy**; **no per-video cost ceiling** today
CPU droplet ? stub	trajectory hybrid + `detector_impl=stub`+`skip_ai_handoff`	nothing	**MOCK**	not real tracking ? UI must label "stub (CPU mock ? not real detection)"; demo/regression only

CPU droplet ? motion	`segmenter=motion`	nothing	REAL, low-quality	only no-key/no-GPU real segmenter; never surface `server/receiver/events` (fabricated)
GPU host ? native WASB (in-flight)	`wasb`+`detector_impl=native` ? `NativeWasbRunner` (in-proc torch)	a **GPU host ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU**; `WASB_WEIGHTS`	**REAL**, on-device	partially wired (fusion builds it, trajectory refuses); **owner byte-parity gate**; **Cloud Run/GCE need a containerized worker ? no Dockerfile exists, and WASB's torch-1.11 conda env makes the image non-trivial**
Owner Win+WSL ? auto	`wasb`+`detector_impl=auto` (WSL2+conda)	Windows+WSL2+GPU	REAL (today)	Windows-only transport; the working real-GPU path for owner verification
Remote GPU worker via bucket (reserved)	console CPU + `storage=s3`; separate GPU host runs `backend.worker infer`	bucket creds; a GPU host; a **net-new orchestration endpoint**	REAL (remote)	FastAPI loop and worker CLI are **separate** today; "remote GPU as a setting" is **design** ? selector shown **disabled** in MVP

Provider note: **Cloud Run + GPU** (scale-to-zero, pay-per-job) is the strongest fit for the engine's own **per-video-billing** economics (`DECOUPLED_COMPUTE` #246) ? gated on the **containerization feasibility check** above.

4.4 Auth / exposure (non-negotiable invariant)
No engine auth, by design; the gate is external. **Invariant (all three must hold):** engine bound to `127.0.0.1` + firewall closes `8000` + edge gate up. Go-live checklist `[HOSTING_PLAN.md:102-108]`: Access/basicauth ON; **rotate the chat-shared `GEMINI_API_KEY`**; set `security.allowed_video_root`; rate-limit; per-user quota.

4.5 UI operator console (web/ SPA ? superset of RallyPicker)
- **S1 Ingest & Process** ? the source menu (§4.2) + a segmenter picker from `/api/capabilities` ? `POST /api/process`.
- **S2 Progress** ? 3s poll of the free-text stage (`hashing/validating/segmenting/finalizing`), a **"one job at a time"** notice, and the restart-failure terminal state.
- **S3 Rally pick ? reel** ? **embed RallyPicker verbatim** (`PER_RALLY_SELECTION.md`): honor `min_shot_count`; never render motion `server/receiver/events` or a fake confidence meter. **Per-rally is ONE screen of the console.**
- **S4 Reels / History** ? needs `GET /api/videos` + `GET /api/reels/{video_id}` for **reload-survival**.
- **S5 Settings** ? render `config` + `capabilities`; the compute/storage **target selector shown DISABLED** ("single-box: local") until the bucket-worker bridge lands.

4.6 Reuse map
Reused (no re-plumb): the full `/api` loop; the in-process worker; the chunked upload endpoints (un-called); LocalStorage default; the detector seam incl. the landing `NativeWasbRunner`; the **RallyPicker** design + anchors; the `web/` scaffold; the `HOSTING_PLAN` gate; the `SETUP_NEW_MACHINE` runbook. **Net-new:** Caddyfile + systemd units; `GET /api/capabilities`; `GET /api/videos`; `GET /api/reels/{video_id}`; `GET /api/health`; the SPA console screens; ingest-from-bucket wiring; a retention sweep; (reserved) the bucket-worker bridge.

5. Observability ? logging & monitoring `[required, first-class]`
Launch-readiness directive (owner 2026-06-20): **logs and monitoring taken seriously, with a full log-analysis pass across every new code path.** The engine already uses a centralized Python logger `[verified main.py:12-17]` ? carry that discipline into every new path.

- **Correlation ID end-to-end.** Thread the existing `job_id` (from `/api/process`) **plus** an `upload_id`/`request_id` through ingest ? process ? jobs ? compile ? highlights, and into the worker for the bucket path, so **one CUJ is traceable** across UI ? FastAPI ? worker ? GPU host. Log it on every line.
- **Structured logs.** New endpoints/screens emit structured (JSON-able) logs at **entry / result / error** ? not just "it ran". The SPA logs client-side errors + failed fetches with the same correlation id.
- **Loud failures, never silent drops.** Carry the engine's ethos (#240 "surface failed chunks loudly") + RallyPicker's `min_shot_count`-warning rule: a partial/empty reel must **never** look complete; every dropped clip/chunk is logged + surfaced in the UI.
- **Health + metrics.** Ship the missing `GET /api/health` `[HOSTING_PLAN.md:42]` and a thin metrics surface (jobs queued/running/failed, last error, disk free, per-video cost once a

ledger exists) for the engine-mode banner + alerting.
122 - **Aggregation for transient hosts.** A scale-to-zero Cloud Run / GPU host is **ephemeral** ? logs must ship to a durable sink (the bucket or a log service), not die with the instance.
123 - **Log-analysis pass before merge (DoD item).** Every PR that adds a code path includes a check that the path logs meaningfully, and the **logs are inspected during CUJ live-replay** (§8 Layers 1?3). Ties [[feedback-launch-quality-bar]].
124
125 ## 6. Risk table
126 | Risk | Today | Mitigation |
127 |---|---|---|
128 | No app auth + CORS `*` `[main.py:42]` | open if any of {localhost-bind, firewall, gate} slips | enforce all three; **LIVE posture test** that raw `:8000` is refused from outside |
129 | Test ports on shared droplet | MinIO `9000/9001` + GCS emu `9023` | firewall closes them |
130 | Retention | `output/`+`uploads/` originals **never deleted** | sweep / keep-proxy-only before any non-owner exposure |
131 | Surprise Gemini/GPU bill | **no cost ceiling/ledger** | per-video cap; `ai_max_workers` (default 5) tunable; `max_workers=1` job serialization |
132 | Native WASB unverified | partially wired; owner-parity-gated | treat as not-production-verified until owner byte-parity on a real GPU box |
133 | New engine endpoints | `capabilities`/`videos`/`reels`/`health` don't exist | coordinate engine PRs with active sessions; **branch from `origin/master`, re-verify HEAD, explicit `git add`** |
134 | Drive ingest overstated | `gdrive://` is a stub | don't present Drive as live; require Drive?bucket pre-step until real |
135 | No Dockerfile | runbook-only reproducibility | feasibility-check a worker image before promising Cloud Run/GCE |
136 | Shared-checkout collisions | active worker sessions share the engine tree | the branch-safety protocol on every engine-side PR |
137 | **Provisioning-credential leak** | a DO API token / cloud creds could leak if pasted in chat or committed (cf. the **already-rotated Spaces key** exposed in chat, `DECOUPLED_COMPUTE/HANDOFF.md`) | authenticate the CLI **on-box** (`doctl auth init` interactive / `gcloud` already authed) so the secret never enters chat or the repo; least-privilege token, revoke after; creds 0600, never committed; **creating paid GPU infra is a confirm-first action every time** |
138
139 ## 7. Honest limits ? what this does NOT do
140 Does **not** add multi-tenant identity/RLS (single-tenant alpha; uploads/outputs global). Does **not** run real WASB on the CPU droplet (Gemini or stub only). Does **not** make the remote-GPU bucket path work (design; selector disabled). Does **not** provide in-UI source playback, job cancel, or a true progress bar (coarse free-text stage, poll-only). Does **not** containerize (runbook; no Dockerfile `[verified]`). Does **not** verify native WASB (owner byte-parity on a real GPU box, not this offline suite). A green SPA + contract tests prove the **console plumbing**, not engine rally-detection quality (a separate track).
141
142 ## 8. CUJs & live-test plan `[required]`
143 **CUJs:** **CUJ-1** ingest (Drive/bucket/local) ? index ? reel ? download (core); **CUJ-2** re-open a past match after reload; **CUJ-3** pick-by-query (deterministic filters + honesty-hiding); **CUJ-4** settings/compute placement (truthful capabilities).
144
145 **Live-test plan ? 5 layers, every CUJ replayed LIVE throughout development:**
146 - **L1 ? Mock engine (hermetic, fast, default + CI).** A tiny FastAPI/MSW implementing the **exact** contract (upload; `process` 202; `jobs` state machine incl. restart-failure; `query` with a motion fixture for honesty-hiding; `config`; `capabilities` per box profile ? CPU-no-key / CPU-with-key / GPU; `compile`; `highlights` tiny mp4; the new `videos`/`reels`). The SPA drives **all 4 CUJs in <1s**, **replayed via the Claude Preview/browser tools** (start dev server ? fill the ingest picker ? click through process/poll/RallyPicker ? assert the reel link via snapshot/network). Catches reload-survival (CUJ-2), the `play_segments` field-name bug (CUJ-3), capability honesty (CUJ-4).
147 - **L2 ? Engine-contract checks (real FastAPI, GPU-free).** Boot `python -m backend.main --server` with `storage=local` + `detector_impl=stub` + `skip_ai_handoff` and assert the **mock and the real engine agree shape-for-shape** (202 keys, jobs enum, `result.play_segments` schema, compile `{status,filepath}`, highlights content-type). Every new endpoint is born with a mock

handler **and** a real-engine contract test **in the same PR**. Runs in CI.
- **L3 ? Real local e2e (the MVP substrate).** Through the gate on the droplet: CUJ-1 with `stub` (key-free) and `gemini` (real, short clip to bound cost); the **exposure-safety posture test** (raw `:8000`/`:900x` refused from outside) as an assertion, not an assumption; reuse the `parity.yml` dual-host pattern.
- **L4 ? CI regression (green gates merge).** `npm test` + a stub-engine contract job + headless browser-replay CUJ scripts; one small PR per tracker row off `origin/master`, **no stacking**; coverage at/above the floor; **logs inspected during replay** (§5).
- **L5 ? Owner hardware e2e (handed off).** Native WASB **byte-parity** on a real GPU box; real-4K **golden-corpus** runs (Gemini + native); Modern-Standby/`cv2`-vs-`ffmpeg` behavior; a multi-GB browser upload. Reported back into §9, stated plainly as **not verified** in the offline suite.

9. Deliverables & progress tracker ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking; CI green gates merge; coverage ? floor; each PR ships its tests + meaningful logs (§5).

ID	Deliverable	Depends on	Gated?	Status	PR
T0	This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers	?	owner	?	(this PR)
T1	Mock engine + capability fixtures + browser-replay CUJ scripts (L1)	T0	no	?	?
T2	Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key (P1)	?	owner	?	?
T3	L3 live posture test (raw `:8000`/`:900x` refused from outside)	T2	no	?	?
T4	S1 ingest (local upload wired) + segmenter picker	T1	no	?	?
T5	S2 progress (poll + one-job + restart-failure)	T4	no	?	?
T6	S3 embed **RallyPicker** + compile + download	T4	no	?	?
T7	*(engine)* `GET /api/capabilities` + `GET /api/health` + banner	T1	coord	?	?
T8	*(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival)	?	coord	?	?
T9	S5 Settings (config+capabilities; disabled compute/storage target)	T7,T8	no	?	?
T10	Observability pass: correlation id end-to-end + structured logs + metrics (§5)	T4	no	?	?
T11	Retention sweep for `output/`+`uploads/`	?	owner	?	?
T12	Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? **engine half DONE** (`/api/process` accepts URIs, M2/[#280], `[verified]`); remaining = the SPA ingest call (= **B-P2**)	T4	coord	?	engine done; SPA pending
T13	**GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner byte-parity (P4)	branch	? owner	?	?
T14	*(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5)	T8,T12	? owner	?	?

Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the start ? owner provisions the host + token; the engine session lands native WASB; the UI MVP does **not** block on it (ships on Gemini).

10. Open questions ? owner / external
Blocking gates: (1) **edge-auth** ? Caddy basicauth (self-contained) vs cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention defaults** ? auto-delete originals after compile? keep proxies+intervals only? window? (privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates with the active worker sessions.
Nice-to-knows: console home (web/ vs engine static); a review-playback streaming endpoint; per-user scoping timing; containerization vs runbook.

11. Definition of done
- [ ] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via

the ingest menu.
181 - [ ] Raw `:8000`/`:900x` unreachable from outside (**tested**, not assumed).
182 - [ ] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped honestly (not faked).
183 - [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
184 - [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via `/api/videos` + `/api/reels`.
185 - [ ] **Observability:** correlation id threaded end-to-end; structured logs at every new path; loud failures; logs inspected during CUJ replay.
186 - [ ] Retention sweep in place; `GEMINI_API_KEY` rotated; `allowed_video_root` set.
187 - [ ] CI green (L2 contract + L4 regression); coverage ? floor; §9 tracker all ?/?-resolved.
188
189 ## 12. References
190 **Engine code `[verified this session]`:** `backend/main.py:12-17,42,52,307-328,331-340,342-396,638`; `backend/api/uploads.py:64,89,123,147`; `backend/api/job_worker.py:37`; `backend/api/static/app.js:88,344,373`; `backend/storage/{base.py,local.py:33,ref.py,__init__.py:43}`; `backend/config/models.py:129,183,298`; `backend/pipeline/segmenters/{trajectory_hybrid.py:57-63,76-85, fusion_hybrid.py:76-86, gemini.py:470-485, motion.py:191-198}`; `backend/pipeline/detectors/{native_wasb_runner.py, stub_runner.py}`; `backend/worker/__main__.py:88,98,149`; `config.json` (`default_segmenter: gemini`). **Engine docs:** `HOSTING_PLAN.md:42,102-108`; `VIDEO_LOCALITY_MODEL.md`; `DESKTOP_APP_PLAN.md`; `SETUP_NEW_MACHINE.md`; `PLATFORM_ARCHITECTURE.md`; `DECOUPLED_COMPUTE/{README,02,04,07,HANDOFF}.md`; `UI_PROPOSAL.md:41-66`; `PROJECT_DOC_TEMPLATE.md`. **This repo:** `PER_RALLY_SELECTION.md`, `REQUEST_ACCESS_FORM.md`, `PORTABILITY.md`, `web/README.md`.
191
192 ---
193 ### Changelog
194 - 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs (FastAPI-native / unified-origin / decoupled-profile) into the reverse-proxied unified-origin spine + compute-as-a-setting framing, via a multi-agent design pass; every engine-fact anchor re-verified against source before writing (catching the native-WASB partial-wiring + `ai_max_workers` default nuances). Folded in the owner's ingest-menu (Drive/bucket/local), multi-provider GPU, and first-class Observability directives.
195

10. user (2026-06-21T13:35:07.780Z)

1 # Friendly & Multilingual ? non-tech-savvy UX + Hindi/Kannada localization
2
3 > **Status:** `DRAFT` ? owner-gated (implementation starts only on explicit approval) · **Owner:** Avi Dullu · **Created:** 2026-06-21 · **Last updated:** 2026-06-21
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE).
5 > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; thread in `NEXT_STEPS.md`; pointer in engine `docs/DOC_STATUS.md`; memory `current-state` + `next-session-ux-localization`.
6 > **Relation to existing docs:** this is the **implementation project** for two engine design docs ? it *adapts & supersedes-in-practice* `badminton-highlight-indexer/docs/I18N_PLAN.md` (localization) and the UX half of `?/docs/UI_PROPOSAL.md` (non-tech-savvy console). Those stay the canonical *design*; this doc carries the *tracked execution* against today's surfaces.
7 > **Honesty note:** each claim tagged `[verified]` (checked against code this session), `[researched]` (needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL font question (D-FONT) is owner/counsel-gated.
8
9 ---
10
11 ## 0. TL;DR
12 A coach or parent in Bangalore should land on KhelSutra, feel at home, set up their camera with confidence, and get highlights ? **in Kannada or Hindi**, with **zero hand-holding**. This project isolates that goal into one tracked workstream-pair: **(A)** make the UI

super friendly for non-tech-savvy users, and **(B)** localize to Hindi + Kannada. Both surfaces (`web/` React SPA + `site/` vanilla marketing) have **zero i18n today** `[verified]`, so the foundation ? shared `locales/{en,kn,hi}` catalogs + a negotiation contract + CI parity ? is the real deliverable; translations follow. **The single most important caveat:** translations are **human-reviewed** (Gemini drafts are *inference only, never training* ? the standing guardrail), and Kannada/Hindi rendering is **blocked on the OFL font decision (D-FONT)**. The cheapest unblocked high-impact first slice is the **SPA i18n scaffold + language switcher (en-only)** ? it locks the shared catalog/negotiation contract and adds the render-test safety net the SPA currently lacks.

1. Problem & goal

Why now. The public site (khelsutra.guru) is live, clean, honest, and tested. The PMF beachhead is **Bangalore badminton academies**; many coaches, parents, and players are far more comfortable in **Kannada** than English, and most are **not highly tech-savvy** `[verified ? design intent]` (`UI_PROPOSAL.md:25,28`, `I18N_PLAN.md:10-13`, `PMF_MARKET_RESEARCH.md:116`, `KHELSUTRA_PRODUCT_OVERVIEW.md:13/50/61`). An English-only, knob-heavy product adds friction at the exact point of first contact.

Concrete outcome ("good" looks like): a Kannada/Hindi-comfortable, phone-first, non-technical coach/parent can ? without help ? (1) understand what the product does and how to record reliably, (2) request access / use the console, (3) pick rallies and get a reel, and (4) share it on WhatsApp, **entirely in their language**.

Read to get current: this doc's §3?§4; the engine design `I18N_PLAN.md` (canonical i18n design) + `UI_PROPOSAL.md` §2 (personas, the 6 alpha screens, "teach recording *before* upload"); `VIDEO_RECORDING_GUIDELINES.md`; this repo's `PER_RALLY_SELECTION.md` (the honest SPA scope) + `REQUEST_ACCESS_FORM.md`.

2. Decisions locked (structural ? owner directive 2026-06-21)

#	Decision	Source / date	Implication
D1	**Isolate as ONE tracked project** spanning both workstreams (A: UX, B: kn/hi l10n)	owner, 2026-06-21	This doc; §7 = source of truth; archived on DoD
D2	**Home = `khelsutra/docs/`** (the implementation repo)	owner-gated tracked-doc convention	`web/` + `site/` both live here; §7 tracks khelsutra PRs; engine P2 backend-l10n is a cross-repo deferred row
D3	**Adapt `I18N_PLAN.md` to today's surfaces:** P1 (vanilla `t()` shim) ? `site/`; P3 (react-i18next) ? `web/`; **drop the dead `backend/api/static` retrofit**	`[verified]` the doc targets `backend/api/static` (`I18N_PLAN.md:25,30,156,191`) which exists only in the engine repo, not here	The doc's framework-agnostic *foundation* (catalog format + negotiation + ICU plurals + CI) survives unchanged; only the wiring re-points
D4	**Engine P2 backend-localization stays owner-gated + cross-repo** (not in this DoD)	`CLAUDE.md` guardrails	Dynamic engine error/report text localizes later; tracked as a deferred row
D5	**Archive trigger = en+kn+hi live on BOTH surfaces** (+ the non-tech UX baseline), then run the engine `DOC_STATUS.md §2` archival checklist	owner, 2026-06-21	See §9
D6	**D-LOCALES: `en` + `kn` for v1, `hi` next; always-fallback to `en`**	owner, 2026-06-21	Kannada-first beachhead; `hi` = catalog-only (both 2-form CLDR)
D7	**D-FONT: OFL-for-media carve-out ? self-host Noto Sans Kannada/Devanagari** (counsel may still confirm)	owner, 2026-06-21	Unblocks B4/B5/A3 + the kn/hi reel watermark; an explicit exception to the MIT/Apache/BSD code guardrail (fonts = media)
D8	**D-SURFACE-FIRST: lead with the `web/` SPA scaffold; SKIP the dead static-UI retrofit**	owner, 2026-06-21 (via B1 go-ahead)	B1 is greenlit; site recording-education localized early
D9	**D-CATALOG-HOME: co-locate `locales/` + CI in `khelsutra`** (shared, consumed by SPA + site)	owner, 2026-06-21 (via B1 go-ahead)	No cross-repo coupling

*Only **D-BACKEND-DEPTH** (§8.3) remains open ? and it's a deferred, cross-repo engine concern (B7), not a v1 blocker.*

3. Foundation ? research / prior art

Current-state facts `[verified]` this session (adversarially checked against the working tree):

- **`web/` SPA: zero i18n.** `web/package.json` deps are only `react`/`react-dom`; no

`i18next`/`react-intl`/`formatjs`; no `locales/` dir; no `t()`/`useTranslation`/`data-i18n` anywhere. User-facing English is hardcoded in `.tsx` (e.g. `QueryBar.tsx:46-47,56,72,76,78`; `App.tsx:115,211,224`) ? ~50 literals + ~16 derived labels.
- **`site/` marketing: vanilla, zero i18n.** Three `<html lang="en">` files (`public/{index,capture,own-model}.html`) with inline CSS, no framework; the only scripts are a CF analytics beacon (`index.html:595`) + one inline form/reel IIFE (`index.html:597-633`). `server.js` hardcodes English responses; no locale handling.
- **Recording-education is *not* greenfield.** The homepage "Record it right" gist (`index.html:324-364`) + the `/capture` checklist (`capture.html`, routed `server.js:51`, guarded `assets.test.js:94`) already ship the user-facing best-practices surface. *Unshipped:* the empirical-threshold framing and the programmatic pre-flight validator (a code change, not a page).
- **Persona is design intent, not built.** kn/hi + zero-knobs are **proposed/owner-gated**; the app is English-only today (`KHELSUTRA_PRODUCT_OVERVIEW.md:61`). It's Kannada-**first**, Hindi-**next** ? not Kannada-only.

Localization prior art `[researched]`:
- **CLDR plural categories:** Hindi = `{one, other}`; Kannada = `{one, other}`. Both are 2-form (same shape as English) ? once ICU MessageFormat is in the catalog, adding `hi`/`kn` plurals is **catalog-only**, no code. (ICU still required because English `count>1?'s':''` hand-rolling ? present today in the SPA ? doesn't generalize.)
- **react-i18next** (web/): `i18next` + `react-i18next` + `i18next-browser-languagedetector` + an ICU/plural post-processor; lazy-load per-locale JSON; persist via `localStorage`.
- **Shared catalog + tiny shim** (site/): the SAME i18next-flavored JSON consumed by a ~30-line `t(key, vars)` shim that walks `data-i18n` attributes ? no build step, no framework, mirrors i18next interpolation/plural rules for the 2-form languages.
- **Noto Sans Kannada / Noto Sans Devanagari** are **OFL-1.1**; OFL permits commercial bundling/redistribution (fonts = media, not code/weights). Self-host (e.g. `@fontsource`) with `font-display: swap` + subsetting. **Gated on D-FONT.**
- **Non-tech-savvy / HCI4D patterns:** icon **+ text** (never icon-only), non-color selection indicators, large touch targets, mobile-first, WhatsApp as the share channel, a prominent persistent language switcher.

4. Design / architecture
Two surfaces, one shared source of truth.

```
locales/                      # shared catalogs ? co-located in khelsutra (D-CATALOG-HOME, see §8)
  en/  common.json            # source of truth ? devs author keys here
  kn/  common.json            # Kannada  (Gemini-draft ? human-review)
  hi/  common.json            # Hindi    (next ? the extensibility proof)

web/   ? react-i18next        # I18N_PLAN P3: provider in main.tsx; useTranslation in components
site/  ? t(key,vars) shim      # I18N_PLAN P1 retargeted: data-i18n attrs over the static HTML
```

- **Negotiation contract (one order, both surfaces):** `persisted pref (localStorage) ? switcher / ?lang=<code> ? navigator/Accept-Language ? "en"`. Supported locales in a single module; anything else falls back to `en`; **a missing key degrades to English, never to a raw key.**
- **Keys, never prose** ? `t("compile.button")`, with ICU for interpolation/plurals: `"{count, plural, one {# rally} other {# rallies}}"`. This replaces the SPA's current hand-rolled `Segment${count>1?'s':''}`.
- **Translation pipeline (guardrail-clean):** dev authors `en` keys ? `scripts/i18n_draft.py`-style Gemini **draft** (inference only, never training; flags every value for review) ? **owner/native-speaker review** ? CI key-parity ? merge. Gemini never touches `en`, never invents keys.
- **CI enforcement (keeps it "by design"):** (1) **key-parity** ? every locale has exactly the `en` key set (drop a stub `ta.json` ? CI lists every gap = the extensibility proof); (2) **no-bare-strings** ? flags user-facing literals not routed through `t()` (curated allowlist to start).

- **Fonts:** bundle Noto Sans Kannada + Devanagari **iff D-FONT = OFL carve-out**; this also unblocks a Kannada/Hindi **reel watermark** (engine `WatermarkConfig.font_path` already takes a path ? config, not code).

Non-tech-savvy UX levers (ranked impact × effort, grounded in personas; surface-tagged):

Lever	Surface	Impact	Effort	Note
Plain-language copy ? strip engine jargon, keep honest caveats	both	high	small	banners + footer carry backend vocabulary today
Localized **WhatsApp share** for the downloaded reel	both	high	small	WhatsApp is *the* channel for this audience
Language switcher persisted via a shared localStorage key	both	high	small	site promises kn/hi but ships no switcher
Localize **recording-education** (capture + gist) kn?hi	site	high	medium	gates input quality; the highest-leverage kn content
Mobile-first touch targets; tooltip-text ? visible	web	high	medium	tiny targets + desktop layout + invisible tooltips on phones
Icon **+ text** + non-color selection indicator	web	medium	small	never icon-only
Friendly engine-error mapping over raw FastAPI `detail`	web	medium	medium	`CompileResult` passes engine text untranslated
Plain-language framing for the Request-Access GPU/storage Qs	site	medium	small	"GPU"/"S3/GCS" are intimidating
Background-processing reassurance + status as icon chips	web	medium	small	honest expectations on slow uplinks
Fix English-source recording gaps (a diagram) before translating	site	medium	medium	the diagram is reusable across all 3 languages
Tap-only filter controls alongside the English query bar	web	medium	large	the `parseQuery()` parser is English-grammar only
Keep the honest *today-vs-roadmap* split through translation	both	medium	small	never soften "experimental"/"not available yet"

Reuse map (what's new vs reused): *Reused* ? the i18next-flavored JSON + negotiation contract + ICU plurals + key-parity/no-bare-strings CI + draft?review pipeline (all from `I18N_PLAN.md`); the shipped recording pages; the SPA's `lib/format.ts`/`lib/query.ts`; `site/src/assets.test.js` + the SPA's vitest harness. *New* ? `locales/` catalogs; the SPA react-i18next provider + switcher; the site `t()` shim + `data-i18n` wiring; Noto font bundle (gated); a jsdom render-test env for the SPA.

5. Privacy / scope note
i18n touches **presentation only** ? no new personal-data collection, no biometrics, minors-excluded posture unchanged. WhatsApp share opens the user's own client with a pre-filled message; it sends nothing server-side. Gemini-draft translation is inference, not training (no user data in the loop).

6. Honest limits ? what this does NOT do
- A green CI / DONE status proves **key-parity + render + no-bare-strings**, **not** translation *quality* ? that's human-gated and owner-paced.
- Kannada/Hindi will **not render** until the Noto fonts land (D-FONT); until then kn/hi catalogs exist but display tofu/fallback.
- The SPA is **job-tethered**; localizing **dynamic engine errors/report text** needs engine P2 (owner-gated, cross-repo, D-BACKEND-DEPTH) ? out of this DoD.
- The query bar stays **English-grammar parsing**; kn/hi users rely on tap-only controls (a later, larger slice), not free-text Kannada queries.
- **Mobile-*perfect*** layout is an alpha non-goal (usable-on-phone is in scope; pixel-perfect is not).

6.5 Testing strategy ? automatic language negotiation (rigorous, owner-required)
The bar (owner, #41): a user whose **computer/browser language is Kannada or Hindi must get the respective version automatically** ? that is the core promise and it is **proven by tests at every layer, never assumed.** Auto-selection IS the negotiation order (persisted pref ? `?lang=` ? browser/`Accept-Language` ? `en`); each step gets a test.

Negotiation matrix ? every row is an automated test:

#	Signal under test	Setup	Expected	Proven in
N1	**Browser/OS = Kannada**	no persisted pref, no `?lang`, `navigator.languages=["kn-IN", ?]`	resolves to **kn** ? Kannada content (once the kn catalog ships)	**B1** detection · **B5** content
N2	**Browser/OS = Hindi**	`navigator.languages=["hi-IN", ?]`	resolves to **hi** ? Hindi content	**B6** (hi added to the supported set + catalog)
N3	**`?lang=` override**	`?lang=kn`, browser = en	kn	B1
N4	**Persisted pref wins**	`localStorage=kn`, browser = en, no `?lang`	kn (a returning user keeps their choice)	B1
N5	**Region subtag normalizes**	`navigator.language="kn-IN"`	base **kn** (not a "kn-IN" miss)	B1
N6	**Unsupported language**	`navigator="ta-IN"` (Tamil, not yet supported)	**en** fallback ? never a locale we can't serve	B1
N7	**Missing key in active locale**	kn active, a key absent from `kn/common.json`	**en** fallback for that key (no raw key shown)	B1/B5
N8	**Static site (vanilla shim)**	`Accept-Language: kn` / `?lang=kn` on a `site/` page	kn-rendered page (same order as the SPA)	B3
N9	**Backend (engine P2)**	`Accept-Language: kn` to the API	kn customer/report text	B7 (owner-gated)

Levels & tooling:
- **SPA (`web/`):** jsdom tests stub `navigator.language`/`navigator.languages`, `localStorage`, and the `?lang` query ? assert `i18n.language`/`resolvedLanguage` + rendered text per the matrix. *(B1 lands N1/N3?N7 at the **detection** level now ? `kn` is supported; the kn/hi **content** assertions ride the catalogs in B5/B6.)*
- **Static site (B3):** the shared `t()` shim's negotiation is unit-tested the same way; plus a server test that `Accept-Language: kn`/`?lang=kn` returns the kn-rendered page.
- **CI key-parity:** guarantees every locale has every key, so a correctly-negotiated locale can never fall through to a raw key.
- **End-to-end (manual, owner machine ? per I18N_PLAN §6):** set the real OS/browser to Kannada, then Hindi ? load both surfaces ? confirm the right language renders **and the Noto fonts display** (the cloud CI container can't render fonts; the owner eyeball is the final gate).
- **? Query parsing / semantic understanding (forward-looking, owner directive 2026-06-21) ? i18n testing is MANDATORY there.** The query bar is **English-grammar-only** today (a documented limit, tracker A9): the user types English queries (`over 10s`, `top 15 longest`) and a localized **sample-query prompt** shows them what to type in *every* locale (kn/hi included). The **moment** the parser gains **semantic understanding** or accepts **Kannada/Hindi query input** (e.g. typing a Kannada/Hindi phrase, or a conversational/NL builder ? I18N_PLAN P4), i18n tests become a **merge gate**: a kn/hi query (typed or spoken) must parse to the **correct selection**, the localized **parse-feedback** ("parsed-pill") must render in the active locale, and the unsupported/empty-result messages must localize. **No query-NL / semantic-parsing work merges without those tests.**

A locale that's reachable by the switcher but **not auto-selected from the browser language is a bug, not a "later".** This matrix is a DoD gate (§9).

7. Deliverables & progress tracker   ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking.

ID	Deliverable	Workstream	Depends on	Gated?	Status	PR
P0	This tracked project doc + inbound pointers	?	?	No	?	[#41](https://github.com/avidullu/khelsutra/pull/41) · engine [#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275)
B1	**SPA i18n scaffold + language switcher, en-only** (react-i18next + detector + `en` catalog + switcher + persistence + **jsdom render tests** + **browser-language negotiation tests** §6.5 N1/N3?N7)	B	P0	No	?	[#42](https://github.com/avidullu/khelsutra/pull/42)
B2	Extract SPA strings ? `en` catalog (~50 literals + ~16 derived; ICU plural keys); add **key-parity + no-bare-strings CI**	B	B1	No	?	

[#45](https://github.com/avidullu/khelsutra/pull/45) |
132 | A1 | Plain-language copy pass on the `en` catalog (de-jargon; icon+text; non-color selection) | A | B2 | No | ? | ? |
133 | B3 | Tiny shared `t()` shim + `data-i18n` plumbing for `site/`, en-only (proves shared catalogs across surfaces) | B | B2 | D-CATALOG-HOME | ? | ? |
134 | A2 | Localized **WhatsApp share** for the reel (SPA + site) | A | B1/B3 | No | ? | ? |
135 | A6 | Plain-language framing for Request-Access GPU/storage questions | A | B3 | No | ? | ? |
136 | A7 | Fix English-source recording gaps (add a placement diagram) before localizing | A | ? | No | ? | ? |
137 | B4 | **Noto Sans Kannada + Devanagari** bundle (OFL carve-out) ? *SPA done; reel watermark `font_path` = engine follow-up* | B | ? | D-FONT ? | ? | [#46](https://github.com/avidullu/khelsutra/pull/46) |
138 | B5 | **Kannada (`kn`) catalog** ? render ? *SPA done (AI-draft, native review pending); `site/` rides B3* | B | B2/B4 | D-LOCALES ? | ? | [#47](https://github.com/avidullu/khelsutra/pull/47) |
139 | A3 | Localize **recording-education** (capture + homepage gist) to kn | A | B3/B5 | D-LOCALES+D-FONT | ? | ? |
140 | A4 | Mobile-first touch targets + tooltip-text ? visible (SPA) | A | B1 | No | ? | ? |
141 | A5 | Friendly engine-error mapping over raw FastAPI `detail` (SPA, client-side map) | A | B2 | No | ? | ? |
142 | B6 | **Hindi (`hi`) catalog** ? render (the extensibility proof ? just a catalog) ? *SPA done (AI-draft, native review pending)* | B | B5 | D-LOCALES ? | ? | [#48](https://github.com/avidullu/khelsutra/pull/48) |
143 | A8 | Background-processing reassurance + verdict/status as icon chips (SPA) | A | B2 | No | ? | ? |
144 | A9 | Tap-only filter controls alongside the English query bar (SPA) | A | B2 | No | ? | ? |
145 | B7 | *(deferred, cross-repo)* Engine **P2 backend customer-facing localization** (report `customer_*` + rejection reasons) | B | engine | **D-BACKEND-DEPTH, owner-gated** | ? | ? |
146
147 **? MILESTONE ? SPA fully trilingual (2026-06-21):** the `web/` Studio SPA now speaks **en + kn + hi** end to end (B1?B2?B4?B5?B6, all merged). A **Kannada-default browser auto-renders Kannada**, and the switcher walks kn ? hi ? en ? **proven by a Playwright screencast** with `locale: kn-IN` (`khelsutra-i18n-screencast.mp4`; frames captured per locale + `<html lang>` tracked). This meets the owner's success criterion **on the SPA surface**. **Still open for the full §9 DoD:** **B3** (the `site/` marketing surface via the shared `t()` shim ? the homepage/capture pages are still English-only), **A3** (recording-education in kn/hi), the **non-tech UX A-items**, and a **native-speaker review** of the AI-drafted kn/hi catalogs before a wide public launch. Do **not** archive until both surfaces are trilingual.
148
149 ## 8. Open questions ? owner (blocking gates) `[design]`
150 *Status (2026-06-21): 4 of 5 **LOCKED/ADOPTED** by the owner ? see §2 (D6?D9). Only D-BACKEND-DEPTH remains open (and it's deferred, not a v1 blocker).*
151 1. **D-FONT ? ? LOCKED ? (a) OFL-for-media carve-out, self-host Noto Sans Kannada/Devanagari** (counsel may still confirm). An explicit exception to the MIT/Apache/BSD code guardrail (fonts = media). Unblocks B4/B5/A3 + the kn/hi reel watermark.
152 2. **D-LOCALES ? ? LOCKED ? `en` + `kn` for v1, `hi` next, always-fallback to `en`** (both 2-form CLDR ? `hi` is catalog-only).
153 3. **D-BACKEND-DEPTH ? ? OPEN (deferred, not a v1 blocker).** Recommend **customer verdict/report `customer_*` + validation-rejection reasons first; technical/operator text stays English; phase two** (engine repo, owner-gated ? tracker row B7).
154 4. **D-SURFACE-FIRST ? ? ADOPTED ? lead with the SPA scaffold (B1), localize site recording-education early, SKIP the dead static-UI retrofit.**
155 5. **D-CATALOG-HOME ? ? ADOPTED ? co-locate `locales/` + CI in `khelsutra`** (a shared i18n dir consumed by both the SPA and the site). (The engine `I18N_PLAN` assumed the indexer repo; this is the adaptation.)
156
157 ## 9. Definition of done
158 - [ ] **Automatic language negotiation (§6.5) is green** ? a **Kannada browser gets Kannada and a Hindi browser gets Hindi** (N1/N2); persisted-pref / `?lang` / region-subtag / unsupported-fallback / missing-key-fallback all proven (N3?N7) on the SPA, and the static-site shim honors `Accept-Language` + `?lang` (N8).
159 - [ ] **Manual cross-check (owner machine):** OS/browser set to Kannada and to Hindi

each render the respective language on BOTH surfaces, with Noto fonts displaying.
160   - [ ] `en + kn + hi` catalogs live and **rendering** on BOTH `web/` SPA and `site/` marketing (switcher + persistence + `en` fallback; no raw keys shown).
161   - [ ] **Noto Sans Kannada + Devanagari render correctly** ? owner eyeball on a real machine.
162   - [ ] CI **key-parity + no-bare-strings** green; a stub locale flags every gap.
163   - [ ] **Recording-education** (capture + homepage gist) available in `kn` + `hi`.
164   - [ ] **Non-tech UX baseline:** plain-language copy (no raw engine jargon in user-facing strings), mobile-usable touch targets, icon+text, WhatsApp share, honest today-vs-roadmap preserved through translation.
165   - [ ] §7 all rows ? (or explicitly deferred with a reason) ? flip **Status ? DONE** + completion note ? run engine `DOC_STATUS.md §2` archival checklist ? `git mv` to `docs/archives/`.
166
167   ## 10. References
168   **Internal code anchors `[verified]`:** `web/package.json`, `web/src/{App.tsx,main.tsx,components/QueryBar.tsx,components/CompileResult.tsx,lib/query.ts}`; `site/public/{index,capture,own-model}.html`, `site/server.js`, `site/src/assets.test.js`. **This repo's docs:** `docs/PER_RALLY_SELECTION.md`, `docs/REQUEST_ACCESS_FORM.md`, `docs/README.md`, `NEXT_STEPS.md`. **Engine design docs (canonical):** `badminton-highlight-indexer/docs/{I18N_PLAN.md,UI_PROPOSAL.md,VIDEO_RECORDING_GUIDELINES.md,PMF_MARKET_RESEARCH.md,PMF_FIELD_SIGNALS.md,KHELSUTRA_PRODUCT_OVERVIEW.md,DOC_STATUS.md}`. **External `[researched]`:** Unicode CLDR plural rules (kn/hi = one/other); react-i18next + i18next-icu; SIL Open Font License 1.1; Google Noto Sans Kannada/Devanagari.
169
170   ---
171
172   ### Changelog
173   - `2026-06-21` ? created. Isolated the UX + Hindi/Kannada work into one tracked project (owner directive). Grounded in a multi-agent research+verification pass (web/+site/ string maps, Indic-l10n prior art, 5 adversarial verdicts). Decisions D1?D5 locked; 5 product/licensing decisions surfaced as open; §7 tracker seeded with B1 as the recommended first slice.
174   - `2026-06-21` ? owner locked **D6 (en+kn?hi)** + **D7 (OFL Noto carve-out)** and adopted **D8 (SPA-first)** + **D9 (catalogs in khelsutra)**; only D-BACKEND-DEPTH remains open. **B1 greenlit ? ? in progress.**
175   - `2026-06-21` ? **B1 opened ? [#42](https://github.com/avidullu/khelsutra/pull/42)**: SPA i18n scaffold (react-i18next + ICU + detector) + `en` catalog + language switcher + the SPA's first jsdom render tests. Verified typecheck/79-tests/build green + live in-browser (switcher persists, en-fallback). Doc/pointer PRs opened: khelsutra [#41](https://github.com/avidullu/khelsutra/pull/41), engine [#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275).
176   - `2026-06-21` ? **§6.5 Testing strategy added (owner ask, #41):** a rigorous **automatic-negotiation matrix** (N1?N9) so a Kannada/Hindi **browser** auto-gets the respective version ? proven at the detection level now (B1), at the content level with the catalogs (B5/B6), on the static shim (B3), and by a manual font-render eyeball. Wired into the DoD (§9). B1 (#42) gains the detection tests (N1/N3?N7).
177   - `2026-06-21` ? **? SPA MILESTONE: fully trilingual (owner directive "take it to completion + screencast").** B2 (#45, full string extraction + key-parity/no-bare-strings CI) ? B4 (#46, self-host OFL Noto) ? B5 (#47, Kannada) ? B6 (#48, Hindi ? the extensibility proof) all merged. 99 web tests green incl. N1/N2 browser-detection + B5/B6 render + key-parity. **Playwright screencast** (`locale: kn-IN`): a Kannada-default browser auto-renders Kannada ? switch hi ? switch en, each with `<html lang>` tracked (`khelsutra-i18n-screencast.mp4` + per-locale frames). kn/hi are **AI-drafted ? native review pending**. **Remaining for full §9 DoD:** B3 (the `site/` marketing surface), A3 + the UX A-items, native-speaker translation review.
178

11. assistant (2026-06-21T13:35:11.773Z)

Now let me read the ANNOTATE_AND_TRAIN doc and verify the engine surfaces for training and annotation, plus the engine docs that describe the worker/training CUJs.

12. user (2026-06-21T13:35:12.747Z)

1   # Annotate your footage ? train your own model (KhelSutra Studio USP pages)
2

> **Status:** `DRAFT` ? owner-gated on scope · **Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; index in `docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
> **Relation:** peer of `khelsutra/docs/PER_RALLY_SELECTION.md` and `khelsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the latter lands with PR #13); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline. Documents an EXTERNAL public tool (`rally-annotator`, MIT) and the engine's `backend/worker train` CLI; it does not own those repos.
> **Honesty note:** every engine-fact claim is tagged `[verified]` (read against code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not legal advice.

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0. TL;DR
Ship ONE new static **marketing page** (`site/public/own-model.html`, slug `/own-model`) with three anchored sections ? **Annotate your footage · Train your own model · Setup & use** ? telling the bring-your-own-model story and linking the real, public, MIT **rally-annotator** VLC plugin `[verified]`. The single most important caveat: only the **ANNOTATE half is real today** ? the rally-annotator ships (v1.6.4, MIT) and its CSV is directly ingestible by the engine with **no transform** `[verified]` ? while the **TRAIN half is an open, in-build CLI** whose GPU-free path trains a **pipeline-validation mock only**, and whose **real model needs a GPU + an in-flight native WASB runner** behind an **owner-gated ship decision** `[verified gate_verdict harness.py:98]`. The Studio SPA gets a **design-tracked, owner-gated** correction/contribute screen, **never a "Train now" button**. Defensible claim = **ownership / licensing / privacy / consent / minors-excluded**, NOT accuracy-superiority (there is no benchmark vs a paid LLM).

1. Problem & goal
KhelSutra's differentiator vs paid LLMs is that **you can own a model trained on YOUR footage** ? your courts, your camera angles, your busy multi-court halls ? on a commercially-clean, inspectable stack. The concrete, usable entry point already exists: **rally-annotator**, a free MIT VLC Lua extension that turns "watch a match" into "produce golden labels" `[verified README.md:22-24]`. What's missing is the **education/positioning surface** that tells this story and the **honest documentation** of the end-to-end loop. "Good" = a viewer understands the moat, can install the annotator and produce labels today, and is **never misled** into thinking one-click training exists. Read first: `rally-annotator/README.md`, `rally-annotator/docs/CSV_FORMAT.md`, `badminton-highlight-indexer/backend/worker/__main__.py`, `khelsutra/docs/PER_RALLY_SELECTION.md`, `khelsutra/site/public/index.html` (the honesty voice).

2. Decisions locked
#	Decision	Source / date	Implication
D1	Marketing page ships NOW; SPA screen is design-tracked + owner-gated	This synthesis 2026-06-20	Convert surface first; operate surface deferred
D2	ONE marketing page, three anchored sections (Annotate/Train/Setup)	Avoid thin pages; annotator is external	Clone `capture.html` skeleton; no build step
D3	Annotator docs live on marketing; SPA only links out	`[verified]` annotator is a standalone VLC tool	Do not re-author VLC instructions in the SPA
D4	NO "Train now" button/endpoint in the SPA	`[verified]` training is offline `backend/worker train`	SPA's honest role = correction loop ? grow corpus
D5	Frame USP as ownership/licensing/privacy/consent, NOT accuracy	`[verified]` no user-model-vs-LLM benchmark exists	No accuracy-superiority copy anywhere
D6	Paid LLMs never a training source; MIT/Apache/BSD only; minors excluded; opt-in	`[verified CLAUDE.md]` + `harness.py:178,189-192` attestation	Reuse the site's existing minors + opt-in wording (`index.html:342-349`)
D7	Annotator CSV needs NO transform for the engine label path	`[verified]` §4.2 schema bridge	Document the MD5+stem join as the only real prereq

3. Foundation ? research / prior art `[verified this session]`
The annotator (real, shipping). `rally-annotator` is a single-file VLC **Lua extension** (`vlc/rally_annotator.lua`, VERSION 1.6.4), MIT-licensed, public at https://github.com/Khelsutra/rally-annotator. It is button-driven (no listener capabilities; deliberate ? VLC pause-listener is flaky on macOS / broken in 4.0), reads playback time via `vlc.var.get(input,'time')` (micros/1e6?s, lines 201-207), and appends one row per rally to

`<video-stem>.rallies.csv` next to the video with atomic rewrite (save_all 310-335). Targets 5 net-separated racquet sports sharing one error taxonomy. CI runs a Lua 5.1 dialog suite + layout-snapshot golden diff. VLC **3.0.x only** (4.0 changed the Lua API).
30
31 **The train loop (partly real).** The engine's `backend/worker train` (cmd_train, `__main__.py:231`) pulls `labels/` (+ `videos/` for the detector source), builds per-video trajectory CSVs, assembles a manifest, **shells out** to `python -m training.gen0.harness` (the one-way train/serve wall, `training/README.md:9-11`), and pushes a versioned bundle to `weights/<version>/`. Two trajectory sources, identical downstream: **synth** (DEFAULT, GPU-free, a CPU mock that ENCODES the labels ? plumbing-meaningful only, `stub_runner.py:124-149,131-133`) and **detector** (real WASB, M3.5, **GPU-gated, IN-FLIGHT** ? the Linux-native runner does not exist yet, `02:15`, `04:98-113`). The harness fits a ~1M-param LogisticRegression on CPU and emits the artifact bundle (`weights.json` + reviewable `manifest.json`); **serving consumes only the bundle, never a code import** (ADR-008). The **ship-gate (`gate_verdict`, `harness.py:98`) currently refuses a commercial ship** regardless of F1: WASB checkpoint provenance C3 unresolved (commercial ship refused) + F1@tIoU not yet calibrated (#172).
32
33 **Guardrails (verbatim, non-negotiable).** Paid LLMs (Gemini/OpenAI/Anthropic) = inference/serving/fallback ONLY, **never a training data source** (bundle attests `no_paid_llm_training_labels:True`, `harness.py:192`); label source must be human (annotator golden CSVs, `harness.py:178`). Commercial-clean deps only ? **MIT/Apache-2.0/BSD**. Owned base = Qwen3-VL (Apache) primary; Gemma-4 AMBER/bench-only; Gemma 3 rejected. **Minors excluded** from the training pool; per-user training data needs **explicit opt-in** `[verified CLAUDE.md]`.
34
35 ## 4. Design / architecture
36
37 ### 4.1 The annotate ? CSV ? train ? model flow `[verified except where noted]`
38 1. **ANNOTATE** (rally-annotator, VLC): per rally, Mark START (serve-contact) ? Mark END (shuttle-dead) ? Ending reason ? optional shots ? Save Rally ? appends a row to `<video>.rallies.csv`. No compute, no GPU, no LLM.
39 2. **STAGE**: `python -m src.cli.curate import-local --sport badminton --video-path match.mp4 --annotations-path match.rallies.csv`, and/or populate a corpus prefix `videos/ labels/ manifest.json weights/ outputs/<video_id>/`.
40 3. **TRAIN**: default `python -m backend.worker train --corpus-uri ? --out-uri ? --version 0.1.0` (synth, GPU-free CPU mock, ?2 labeled videos); real = add `--trajectory-source detector --segmenter wasb_hybrid` with `indexing.detector_impl=native` (GPU + native WASB + `wasb_badminton_best.pth.tar`, IN-FLIGHT).
41 4. **HARNESS ? BUNDLE**: parse trajectory + load GT ? frame features ? honest LOVO eval ? fit final LR + tune threshold ? ship-gate ? emit `weights.json` + `manifest.json`.
42 5. **USE** `[researched/design, owner-gated]`: if Gen-0 clears the ship-gate A/B, the bundle drives the existing record?upload?reel loop. No per-user serving UI; no Studio Train button.
43
44 ### 4.2 The schema bridge ? annotator CSV ? engine label CSV (MATCHES, no transform)
45 Annotator writes header `rally_number,start_time,end_time,ending_reason,sport,shots_count` (`rally_annotator.lua:128`; `CSV_FORMAT.md:6`), `%.3f` decimal seconds, end>start. The engine reads **by column name** via two readers, both satisfied:
46 - **Default synth path** ? `synth_trajectory_from_labels` uses `csv.DictReader` reading literally `row["start_time"]`/`row["end_time"]` (`stub_runner.py:139`) ? **MATCHES exactly**; extra columns ignored. (This path needs the `*_time` names specifically ? the short `start/end` alias is NOT accepted here.)
47 - **Canonical GT reader** ? `load_gt_intervals` accepts `start`/`end` OR `start_time`/`end_time`, ignores other columns, also accepts headerless positional col0,col1, tolerates decimal-seconds or MM:SS/HH:MM:SS, BOM, blank/`#` lines (`gt_loader.py:14-90`) ? **MATCHES** the `*_time` branch.
48
49 The worker's own inference CSV uses `start,end,shot_count,ending_reason` (`__main__.py:62-74`) ? order/short-name differ, but order never matters and `gt_loader` accepts both aliases; the annotator's `*_time` form is the **safer** of the two. **The ONE real prerequisite is an IDENTITY/JOIN constraint, not a schema gap:** training keys video?label by **file MD5** and stem-matches `labels/<name>.rallies.csv` ? `videos/<name>.<ext>` (`__main__.py:271-280`; `02:48-60`). Stems must agree and the staged video bytes must be **byte-identical** to what was annotated ? any re-encode changes the MD5 and breaks the join (collector

manifest currently omits `md5`, M2 prereq, `02:62-68`).

4.3 Reuse map
- **Marketing** `[verified]`: clone `site/public/capture.html` skeleton + brand tokens (`index.html:25-32`); reuse `.honest`/`.nots` components (`:152-158`, used `:351-354`, `:400-415`); add a `/own-model` alias in `site/server.js:49-51` (Worker `env.ASSETS.fetch` needs no change, `site/src/index.js:84`); cross-link nav `:218-223`, the "gets sharper" section `:327-356`, footer `:528`; screenshots under `site/public/img/` (precedent `site/og.png` + `make_og.py`).
- **SPA** `[design, gated]`: today `web/src/App.tsx` is one static screen, **no router** (`web/package.json`). A correction screen needs react-router-dom (or a hash switch) + the typed client `web/src/api/client.ts`/`types.ts` + `web/src/lib/log.ts`; it realizes P10 in `PER_RALLY_SELECTION.md:122` and depends on a **new** engine endpoint `PATCH /api/segments/{id}` (dead columns `candidate_segments.human_label/notes/confirmed_segment_id`).
- **Annotator content** `[verified]`: embed CSV example + install commands + ending-reason table **verbatim** from the repo; link, don't re-author.

5. Threat model / privacy
- **Footage stays the user's**: the annotator runs locally in the user's own VLC and writes a CSV next to the video ? labeling happens with **footage never leaving the machine** `[verified]`.
- **Training use = explicit opt-in, never automatic** `[verified CLAUDE.md]`; reuse the site's existing opt-in wording.
- **Minors excluded from any training pool** `[verified CLAUDE.md + live site "Minors are protected"]` ? keep prominent (audience = academies with children).
- **No paid-LLM laundering**: the model never trains on a paid LLM's output (terms/copyright), attested in the bundle (`harness.py:192`).
- **Residual**: base-model (Qwen3-VL) pretraining provenance is unauditable `[researched]` ? owner risk-acceptance required before VLM "train your own model" copy goes beyond roadmap.

6. Honest limits ? what this does NOT do
A shipped marketing page does NOT mean training is one-click (it is an offline CLI), does NOT mean a real model exists (synth = mock; real = GPU + in-flight native WASB + owner-gated ship-gate), and does NOT imply any accuracy claim vs a paid LLM (no benchmark exists). The annotator is a **separate VLC 3.0.x desktop plugin**, not in-browser; it is **click-driven** (no marking hotkeys). The SPA correction screen is **design-tracked + owner-gated** and depends on an unbuilt `PATCH /api/segments` endpoint. Two CSV schemas exist (annotator lighter vs Roora vendor) ? do not conflate.

7. Deliverables & progress tracker ? source of truth
Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, branched from `origin/master`, no stacking.

ID	Deliverable	Depends on	Gated?	Status	PR
P0	This tracked doc in `khelsutra/docs/`, indexed in `docs/README.md`	?	No	?	[#15](https://github.com/avidullu/khelsutra/pull/15)
P1	`site/public/own-model.html` (clone capture.html; 3 sections; honesty fences; placeholder shots)	P0	No	?	(this PR)
P2	`/own-model` alias in `site/server.js` (Worker needs no change)	P1	No	?	(this PR)
P3	Cross-links: nav + footer (the "gets sharper" section link is a nice-to-have, deferred)	P1	No	?	(this PR)
P4	Embed verbatim repo text (CSV/install/ending-reasons) + GitHub link	P1	No	?	(this PR)
P5	Capture + commit real screenshots (consented, no minors, faces redacted); replace placeholders	P1	? Owner (footage)	?	?
P6	Owner: slug + 1-vs-3-page + train-positioning honesty	?	? Owner	?	?
P7	[design] SPA correction/contribute route (RallyReview + link-out)	P8	? Owner	?	?
P8	[design] Engine `PATCH /api/segments/{id}` (M4 corpus extension)	?	? Owner (engine)	?	?
P9	[researched] Serve a shipped Gen-0 bundle into record?reel	native WASB + ship-gate	? Owner+blocked	?	?

81
82 ## 8. Open questions ? owner / external
83 **Blocking gates:** (1) Greenlight how prominently to position "train your own model" (CLI + roadmap). (2) Approve the SPA correction screen + the new `PATCH /api/segments/{id}` endpoint. (3) Owner risk-acceptance on Qwen3-VL provenance before VLM copy. (4) Provide consented no-minors footage for the 5 screenshots.
84 **Nice-to-knows:** slug (`/own-model` vs `/byo-model` vs `/annotate`); one page vs three; react-router-dom vs hash switch; cross-link `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` once PR #13 merges.
85
86 ## 9. Definition of done
87 - [ ] `own-model.html` live with Annotate/Train/Setup sections, honesty fences, verbatim repo text, GitHub link, real (non-fabricated) screenshots.
88 - [ ] `/own-model` alias works on both the Node host and the Cloudflare Worker; cross-linked from nav/section/footer.
89 - [ ] Every claim tagged `[verified]`/`[design]`/`[researched]`; no accuracy-superiority claim; no "Train now" button; VLC-3.0.x-only + GPU/native-WASB-gated + owner-gated stated plainly.
90 - [ ] Schema-bridge + MD5/stem-join prerequisite documented; the two CSV schemas kept distinct.
91 - [ ] §7 tracker reflects real shipped state; doc indexed in `docs/README.md`.
92
93 ## 10. References
94 **Annotator `[verified]`:** `rally-annotator/README.md`, `vlc/rally_annotator.lua` (HEADER :128; now_seconds :201-207; marks :608-628; save_all :310-335; resolve_out_path :253-267), `docs/CSV_FORMAT.md`, `docs/ENDING_REASONS.md`, `LICENSE` (MIT), `.github/workflows/ci.yml`.
95 **Engine `[verified]`:** `backend/worker/__main__.py` (cmd_train :231; _write_intervals_csv :62-74; join :271-280; harness shell-out :321-328), `backend/pipeline/detectors/stub_runner.py` (synth :124-149, :131-133, :139), `training/gen0/harness.py` (gate_verdict :98; label_source :178; attestation :189-192), `backend/eval/gt_loader.py` (:14-90), `docs/SETUP_NEW_MACHINE.md`, `docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md`, `docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md`, `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`, `CLAUDE.md`.
96 **KhelSutra `[verified]`:** `site/public/index.html`, `site/public/capture.html`, `site/server.js`, `site/src/index.js`, `web/src/App.tsx`, `web/package.json`, `docs/README.md`, `docs/PER_RALLY_SELECTION.md`.
97
98 ### Changelog
99 - 2026-06-20 ? Initial DRAFT consolidating annotator + train-loop + UI-surface research into one tracked doc (multi-agent design pass; schema-bridge + no-images + train-guardrails re-verified against source before writing).
100

13. user (2026-06-21T13:35:14.389Z)

```
__init__.py
__main__.py
__pycache__
---API---
__init__.py
__pycache__
auth.py
job_worker.py
static
uploads.py
---ANNOTATIONS---
audio
detectors
personalization
segmenters
stitcher
validators
```

14. assistant (2026-06-21T13:35:17.116Z)

Let me verify the key engine API endpoints and worker CLI surfaces directly, to ground the CUJ list in code.

15. user (2026-06-21T13:35:21.375Z)

```
68:@app.get("/", response_class=HTMLResponse)
138:@app.get("/api/config")
142:@app.get("/api/health")
275:@app.post("/api/process", status_code=202)
326:@app.get("/api/jobs/{job_id}/cost")
341:@app.get("/api/videos/{video_id}/cost")
352:@app.get("/api/jobs/{job_id}")
385:@app.get("/api/highlights/{video_id}/{filename}")
409:@app.post("/api/query")
420:@app.post("/api/compile")
```

16. user (2026-06-21T13:35:21.409Z)

```
156:def cmd_infer(args: argparse.Namespace) -> int:
331:def cmd_train(args: argparse.Namespace) -> int:
452:    p.add_argument("-v", "--verbose", action="store_true",
457:    pi.add_argument("--video-uri", required=True, help="local path / local:// / s3:// / gcs://")
458:    pi.add_argument("--out-uri", required=True, help="output prefix (outputs/<video_id>/? is appended)")
459:    pi.add_argument("--segmenter", default=None, help="default: indexing.default_segmenter")
460:    pi.add_argument("--config", default=None, help="config.json path (default: ./config.json)")
461:    pi.add_argument("--scratch", default=None, help="local scratch dir (default: a temp dir)")
462:    pi.add_argument("--max-frames", type=int, default=None)
463:    pi.add_argument("--done-uri", default=None,
467:    pi.add_argument("--set", action="append", default=[], metavar="k=v",
469:    pi.set_defaults(func=cmd_infer)
472:    pt.add_argument("--corpus-uri", required=True, help="prefix containing labels/ (+ videos/ for "
474:    pt.add_argument("--out-uri", required=True, help="output prefix (weights/<version>/ is appended)")
475:    pt.add_argument("--version", required=True, help="semver for the artifact bundle")
476:    pt.add_argument("--trajectory-source", choices=["synth", "detector"], default="synth",
480:    pt.add_argument("--segmenter", default="wasb_hybrid",
482:    pt.add_argument("--floor", default=None, help="floor baseline JSON (local path or storage URI)")
483:    pt.add_argument("--config", default=None)
484:    pt.add_argument("--scratch", default=None)
485:    pt.add_argument("--fps", type=float, default=30.0, help="synth trajectory fps (detector probes the video)")
486:    pt.add_argument("--frame-width", type=float, default=1280.0,
488:    pt.add_argument("--set", action="append", default=[], metavar="k=v")
489:    pt.set_defaults(func=cmd_train)
```

17. assistant (2026-06-21T13:35:25.474Z)

Let me verify the segmenter options in the CLI and the auth.py surface (M9), plus check the static UI status and entrypoints.

18. user (2026-06-21T13:35:26.465Z)

```
        except StartupCheckError as e:
            logger.error("[STARTUP] refusing to start: %s", e)
            sys.exit(1)


    def main():
```

```
    parser = argparse.ArgumentParser(description="Sports Highlight Indexer Orchestrator")
    parser.add_argument("--video-path", help="Local path to the MP4 sports video file")
    parser.add_argument("--output-dir", default="output", help="Directory where processed clips/highlights are saved")
    parser.add_argument("--setup", action="store_true", help="Launch dependency checker and setup utility")
    parser.add_argument("--debug-clip", type=float, help="Timestamp (in seconds) of a video clip to run detailed diagnostics on")
    parser.add_argument("--server", action="store_true", help="Run the FastAPI local API server")
    parser.add_argument("--port", type=int, default=8000, help="Port to run the FastAPI server on")
    parser.add_argument("--max-frames", type=int, help="Limit processing to the first N sub-sampled frames for debugging")

    available_segmenters = list(segmenter_registry.list_available().keys())
    parser.add_argument("--segmenter", choices=available_segmenters, help=f"Choose rally detection segmenter. Available: {available_segmenters}")
    parser.add_argument("--trim-start", type=float, help="Start time in seconds for the debug trim segment feature")
    parser.add_argument("--trim-end", type=float, help="End time in seconds for the debug trim segment feature")
    parser.add_argument("--trim-output", help="Output filename for the debug trimmed segment (saves in output-dir unless absolute path)")

    args = parser.parse_args()

    if args.setup:
        # Invoke the diagnostics/bootstrap script (renamed from the old root
        # setup.py to scripts/doctor.py so it no longer shadows the PEP 517
        # build system). Resolve its path from this file so `indexer --setup`
        # works regardless of the caller's cwd.
        import subprocess
        doctor_path = os.path.join(
            os.path.dirname(os.path.dirname(os.path.abspath(__file__))),
            "scripts", "doctor.py",
        )
        print(f"[INFO] Invoking diagnostics: {doctor_path} ...")
        subprocess.run([sys.executable, doctor_path])
        return

    if args.trim_start is not None and args.trim_end is not None:
        if not args.video_path:
            print("[ERROR] Please provide --video-path to extract a segment.")
            return

        # Determine output path
        out_filename = args.trim_output or "trimmed_segment.mp4"
        if os.path.isabs(out_filename):
            out_path = out_filename
            out_dir = os.path.dirname(out_path)
            out_name = os.path.basename(out_path)
        else:
            out_dir = args.output_dir
            out_path = os.path.join(out_dir, out_filename)
            out_name = out_filename

        os.makedirs(out_dir, exist_ok=True)
        print(f"[CLI] Extracting debug segment from {args.trim_start}s to {args.trim_end}s from {args.video_path}...")

        # global_config isn't initialized this early; load the typed config so the trim
        # clip carries the same configured stitching paddings + watermark as a real
        # highlight (watermark default-on).
```

```
            stitching = load_app_config().stitching
            compiler = VideoCompiler(out_dir, begin_padding=stitching.begin_padding,
                                     end_padding=stitching.end_padding, watermark=stitching.watermark)
            try:
                res_path = compiler.compile_highlights(args.video_path, [(args.trim_start, args.trim_end)], out_name)
                print(f"[SUCCESS] Trimmed segment saved to: {res_path}")
            except Exception as e:
                print(f"[ERROR] Failed to compile trimmed segment: {e}")
            return

        if args.debug_clip is not None:
            if not args.video_path:
                print("[ERROR] Please provide --video-path to debug a specific clip.")
                return
            run_cli_diagnose_clip(args.video_path, args.debug_clip)
            return

        # Initialize Global Config and DB
        global global_config, db_instance
        global_config = load_app_config()  # returns typed AppConfig
        _enforce_startup_or_exit(global_config)  # M5: fail fast on an incoherent ACTIVE profile

        # The CLI --output-dir overrides the configured output directory. It now lives
        # on the typed model (OutputConfig.output_dir), so every consumer ? including
        # /api/compile ? reads the same value instead of a write-only runtime hack.
        global_config.output.output_dir = args.output_dir
        os.makedirs(global_config.output.output_dir, exist_ok=True)

        db_path = os.path.join(global_config.output.output_dir, global_config.output.db_name)
        db_instance = Database(db_path)

```

19. user (2026-06-21T13:35:26.828Z)

```
"""Cloudflare Access edge-auth ? verify the Cf-Access JWT, derive the tenant ``user_id`` (M9, D9).

**DEFAULT-OFF:** when ``security.edge_auth_enabled`` is False the resolver is a strict NO-OP (returns
``None`` = the local single-user) ? nothing about the existing local path changes. When enabled, the
``Cf-Access-Jwt-Assertion`` header is **required** and **verified** (RS256) against the team JWKS ?
signature + exp + issuer + audience ? so a direct hit to the Cloud Run URL **can't spoof** the
identity header (the explicit D9 anti-spoof requirement). Returns the verified ``user_id`` (the
token's ``email``, else ``sub``).

``jwt`` (``PyJWT[crypto]``) is **lazy-imported** so the base runtime works without it (install the
``auth`` extra for hosted/multi-tenant). The JWKS fetch is **injectable**, so the whole verifier is
unit-tested **offline** with an in-test RSA keypair + a fake JWKS ? no network, no Cloudflare.
"""

from __future__ import annotations

import json
import time
```

```
import urllib.request
from typing import Any, Callable, Mapping, Optional

# Cloudflare Access sends the signed assertion in this header (it terminates at the CF edge).
CF_ACCESS_HEADER = "Cf-Access-Jwt-Assertion"
_JWKS_PATH = "/cdn-cgi/access/certs"
_JWKS_TTL_SEC = 600.0  # re-fetch the team JWKS at most every 10 min (keys rotate slowly)

JwksFetcher = Callable[[str], dict]


class EdgeAuthError(Exception):
    """A required Cf-Access token was missing or failed verification ? the caller returns 401/403."""


# module-level JWKS cache: team_domain -> (fetched_at, jwks_dict). Production only; tests inject.
_jwks_cache: dict[str, tuple[float, dict]] = {}


def _issuer(team_domain: str) -> str:
    return team_domain.rstrip("/")


def _default_jwks_fetcher(team_domain: str) -> dict:
    """Fetch the team's public JWKS (``<team_domain>/cdn-cgi/access/certs``). Network ? only the
    production path reaches here; tests pass an injected fetcher."""
    url = _issuer(team_domain) + _JWKS_PATH
    with urllib.request.urlopen(url, timeout=5) as resp:  # noqa: S310 (https team domain)
        data: Any = json.load(resp)
    if not isinstance(data, dict) or "keys" not in data:
        raise EdgeAuthError(f"unexpected JWKS shape from {url}")
    return data


def _get_jwks(team_domain: str, fetcher: Optional[JwksFetcher]) -> dict:
    """Return the JWKS for ``team_domain``, cached with a TTL. An injected ``fetcher`` (tests)
    bypasses the cache so each test is deterministic."""
    if fetcher is not None:
        return fetcher(team_domain)
    now = time.time()
    cached = _jwks_cache.get(team_domain)
    if cached and (now - cached[0]) < _JWKS_TTL_SEC:
        return cached[1]
    jwks = _default_jwks_fetcher(team_domain)
    _jwks_cache[team_domain] = (now, jwks)
    return jwks


def verify_cf_access_token(token: str, *, team_domain: str, aud: str, jwks: dict) -> str:
    """Verify a Cf-Access JWT against ``jwks`` (RS256; checks sig + exp + issuer + audience) and
    return the user_id (``email`` else ``sub``). Raises ``EdgeAuthError`` on ANY failure."""
    try:
        import jwt
        from jwt import PyJWKSet
    except Exception as exc:  # pragma: no cover - only when the auth extra isn't
```

```
installed
        raise EdgeAuthError(
            "PyJWT[crypto] is required for edge auth ? install the 'auth' extra "
            "(pip install -e .[auth]) on a host with edge_auth_enabled."
        ) from exc

    try:
        kid = jwt.get_unverified_header(token).get("kid")
        if not kid:
            raise EdgeAuthError("token header has no kid")
        signing_key = PyJWKSet.from_dict(jwks)[kid]
        claims = jwt.decode(
            token, signing_key.key, algorithms=["RS256"],
            audience=aud, issuer=_issuer(team_domain),
            options={"require": ["exp", "iss", "aud"]},
        )
    except EdgeAuthError:
        raise
    except KeyError as exc:  # kid not in the JWKS
        raise EdgeAuthError(f"no JWKS key for kid {exc}") from exc
    except Exception as exc:  # jwt.* errors: bad sig / expired / wrong aud|iss / malformed
        raise EdgeAuthError(f"Cf-Access token verification failed: {exc}") from exc

    user = claims.get("email") or claims.get("sub")
    if not user:
        raise EdgeAuthError("verified token carries no email/sub claim")
    return str(user)


def resolve_tenant(headers: Mapping[str, str], config: Any, *,
                   jwks_fetcher: Optional[JwksFetcher] = None) -> Optional[str]:
    """Resolve the tenant ``user_id`` for a request, or ``None`` when edge auth is OFF.

    DEFAULT-OFF (``security.edge_auth_enabled=False``) ? ``None`` (the local single-user; unchanged).
    Enabled ? require + verify the Cf-Access header; raise ``EdgeAuthError`` (? 401/403) on any miss.
    ``jwks_fetcher`` is injectable for offline tests.
    """
    sec = getattr(config, "security", None)
    if not sec or not getattr(sec, "edge_auth_enabled", False):
        return None  # DEFAULT-OFF ? no tenant identity, byte-identical to today

    team_domain = getattr(sec, "cf_team_domain", "") or ""
    aud = getattr(sec, "cf_access_aud", "") or ""
    if not team_domain or not aud:
        raise EdgeAuthError(
            "edge_auth_enabled but security.cf_team_domain / cf_access_aud are not configured")

    # Headers are case-insensitive over HTTP; Starlette's Headers handles that, but support a plain
    # dict too (tests) by trying both casings.
    token = headers.get(CF_ACCESS_HEADER) or headers.get(CF_ACCESS_HEADER.lower())
    if not token:
        raise EdgeAuthError(f"missing {CF_ACCESS_HEADER} header (request did not pass Cloudflare Access)")

    jwks = _get_jwks(team_domain, jwks_fetcher)
    return verify_cf_access_token(token, team_domain=team_domain, aud=aud, jwks=jwks)

```

20. assistant (2026-06-21T13:35:30.438Z)

Now let me check the rally-annotator repo status and the engine's training README to confirm the train CUJ surfaces, plus verify the static UI is dead vs live.

21. user (2026-06-21T13:35:33.693Z)

```
=== rally-annotator HEAD ===
d5d59f2 v1.6.4: guard unsaved rallies from silent loss + flag edit mode on Mark buttons (#13)
=== files ===
CHANGELOG.md
LICENSE
README.md
docs
test
vlc
=== version ===
    - The version now shows next to the plugin name in VLC's "Active Extensions"
      VLC shows the descriptor title verbatim, so the version is baked into it.
local VERSION = "1.6.4"
==== ENGINE static app.js exists? ====
app.js
index.html
style.css
```

22. user (2026-06-21T13:35:34.260Z)

```
@app.get("/", response_class=HTMLResponse)
def serve_index():
    index_path = "backend/api/static/index.html"
    if os.path.exists(index_path):
        with open(index_path, "r", encoding="utf-8") as f:
            return f.read()
    return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in backend/api/static/</h3>"

# Global variables initialized at startup
db_instance: Optional[Database] = None
global_config: Optional[AppConfig] = None  # now typed (Pydantic model)

# Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTION_POLICY.md P3)
# now lives in backend.api.job_worker. Re-exported here so the names stay addressable on
# `backend.main` (`JobWaiting`, `_drain_due_jobs`, `_arm_wake_timer`, `_get_executor`,
# `_run_claimed_job`, `_MAX_DRAIN_WAKE_SEC`, ...). The worker resolves these seams THROUGH
# this module object at call-time, so `monkeypatch.setattr(backend.main, ...)` still
# intercepts the inter-function calls (a parked job arms the PATCHED `_arm_wake_timer`,
# so no real daemon timer leaks in tests). Behavior is unchanged by the extraction.
from backend.api.job_worker import (  # noqa: F401  (re-exports: names stay addressable on backend.main)
    JobWaiting,
    _arm_wake_timer,
    _drain_due_jobs,
    _get_executor,
    _job_to_request,
    _MAX_DRAIN_WAKE_SEC,
    _maybe_record_job_cost,
    _run_claimed_job,
    _schedule_next_drain_if_waiting,
)

# Legacy thin wrapper for any remaining direct calls during transition.
# New code should import load_app_config / AppConfig from backend.config directly.
def load_config(config_path: str = "config.json") -> Dict[str, Any]:
    cfg = load_app_config(config_path)
    return cfg.model_dump(mode="json")
```

```
# --- API Models ---
class ProcessRequest(BaseModel):
    video_path: str
    player_pool: Optional[List[str]] = None
    max_frames: Optional[int] = None
    segmenter: Optional[str] = None

class QueryRequest(BaseModel):
    video_id: str
    server: Optional[str] = None
    receiver: Optional[str] = None
    duration_min: Optional[float] = None
    event_type: Optional[str] = None

class CompileRequest(BaseModel):
    video_id: str
    segment_ids: List[int]
    output_filename: str

def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
    """Commit a (re)segmentation with destroy-AFTER-confirm semantics.

    On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
    fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark) and
    keep the prior good results intact ? a failed/empty re-run never destroys prior data.
    """
    if segments:
        db_instance.prune_segments(video_id, play_upto=play_wm, cand_upto=cand_wm)
    else:
        db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)


# --- API Endpoints ---
@app.get("/api/config")
def get_config():
    return global_config

@app.get("/api/health")
```

23. assistant (2026-06-21T13:35:38.269Z)

I now have enough verified grounding. Let me do two final checks: confirm the i18n screencast/CUJ artifacts and the worker train detail, plus confirm the segmenter registry options available.

24. user (2026-06-21T13:35:41.911Z)

```
=== segmenter registrations ===
backend/pipeline/segmenters/base.py:39:    def register(self, segmenter_cls: Type[BasePlaySegmenter]):
backend/pipeline/segmenters/base.py:52:        """Returns a dict of registered segmenter names and their descriptions."""
backend/pipeline/segmenters/fusion_hybrid.py:189:segmenter_registry.register(FusionSegmenter)
backend/pipeline/segmenters/gemini.py:564:segmenter_registry.register(GeminiPlaySegmenter)
backend/pipeline/segmenters/motion.py:255:segmenter_registry.register(MotionPlaySegmenter)
backend/pipeline/segmenters/tracknet_hybrid.py:42:segmenter_registry.register(TrackNetHybridSegmenter)
backend/pipeline/segmenters/trajectory_hybrid.py:18:This base is deliberately NOT registered; only concrete subclasses register.
backend/pipeline/segmenters/wasb_hybrid.py:49:segmenter_registry.register(WasbHybridSegmenter)
backend/pipeline/segmenters/yolo_hybrid.py:438:segmenter_registry.register(HybridYoloSegmenter)
=== khelsutra web tests count proof ===
```

```
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3
?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
CompileResult.tsx
IngestPanel.test.tsx
IngestPanel.tsx
LanguageSwitcher.tsx
QueryBar.examples.test.ts
QueryBar.tsx
RallyGrid.tsx
SelectionFooter.tsx
```

25. user (2026-06-21T13:35:43.114Z)

```
No matches found
```

26. assistant (2026-06-21T13:35:48.251Z)

I have thoroughly verified the engine and UI surfaces. Note the IngestPanel.tsx was merged to khelsutra master (#56) ? the in-UI bucket ingest is now landed, and DRY_RUN_CUJ §9 confirms the live e2e ran. I have enough grounding to enumerate every completed CUJ.

27. user (2026-06-21T13:37:20.055Z)

Structured output provided successfully

28. assistant (2026-06-21T13:37:25.458Z)

I've enumerated every completed end-to-end CUJ that the packaged app must preserve, grounded in the engine code and the khelsutra docs, and returned them via StructuredOutput. The list spans 9 CUJs (8 DONE, 1 partial) plus the decoupled-serving seam and the job/UI invariants, with verified file:line and doc-path evidence for each, and the packaging implications/risks/open questions for shipping a batteries-included downloadable app.